instituto superior de engenharia de lisboa

Electrical Engineering BSc

Electric Machinery II

Exercises and Industrial Application Problems

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Exercises and Industrial Application Problems

Electric Machinery II

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Preface

This collection of exercises and industrial application problems is organised into three chapters:

- DC machines
- Three-phase synchronous machines
- Electrical machine transients

This document presents exercises of a more conceptual nature alongside industrial application problems, allowing students to progress from fundamental concepts to practical situations encountered in industry. In each chapter, the exercises or problems are numbered sequentially (Exercise 1, Exercise 2, …), with their scope made clear by the context and problem statement itself.

The odd-numbered exercises are fully worked through, while the even-numbered exercises are proposed as work items, with only the numerical results given. The latter are primarily intended to be solved during theoretical-practical classes, adopting Problem-Based Learning as the teaching and learning methodology. In this way, discussion and critical analysis of the topics are encouraged, fostering the development of analytical thinking and problem-solving skills.

Among the worked exercises, some are solved analytically as an introduction to, or consolidation of, key concepts, while others are solved using the scientific computing language `Julia`, through computational documents (*notebooks*) developed using the integrated development environment `Pluto.jl`. The use of interactive *notebooks* allows students to deepen their understanding of fundamental theoretical concepts and to develop essential skills for professional practice. In addition, this exploratory practice allows students to apply their knowledge to solve exercises interactively, to visualise and analyse results graphically, and to gain practical experience with engineering tools.

Lisbon, September 2025

Ricardo Luís

The real process of education should be
the process of learning to think through
the application of real problems.

— **John Dewey** (1859-1952)

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DC MACHINES

Exercise 1

A six-pole DC machine has 360 conductors in armature slots. Each magnetic pole has a pole arc of 20 cm, a depth of 20 cm and a magnetic flux density of 0.8 T. With the rotor running at a speed of 1000 rpm, determine the induced electromotive force, E_0, obtained if the machine has:

a) a lap-wound armature winding;

b) a wave-wound armature winding.

Worked solution:

Let:

- $2p = 6$ poles; $z = 360$ conductors;
- $b = 20$ cm (pole arc); $l = 20$ cm (depth)
- $B = 0.8$ T

Pole face area, A_p:

$$A_p = b\ l = 0.2 \times 0.2 = 0.04 \text{ m}^2$$

The magnetic flux per pole, ϕ_p, is then:

$$\phi_p = B\ A_p = 0.8 \times 0.04 = 0.032 \text{ Wb}$$

With the speed in rpm, the armature constant, k, is given by: $k = \dfrac{z\ p}{60\ a}$

Thus, the induced electromotive force, E_0, in the armature winding is obtained:

a) Lap winding, $a = p$ (number of brushes equal to number of poles):

$$E_0 = k\phi_p n = \frac{360 \times 3}{60 \times 3} \times 0.032 \times 1000 = 192 \text{ V}$$

b) Wave winding, $a = 1$ (only one pair of brushes):

$$E_0 = k\phi_p n = \frac{360 \times 3}{60 \times 1} \times 0.032 \times 1000 = 576 \text{ V}$$

(Source: Adapted from exercise 5.5 in [Guru & Hızıroğlu, 2003])

Exercise 2

Consider the armature of a four-pole DC generator with a lap winding placed in 28 slots with 10 conductors in each. The magnetic flux per pole is 40 mWb and the armature (rotor) speed is 1200 rpm. The generator supplies a load, and the current in each conductor is 2 A. What are the torque and power developed by the generator?

> **Results:**
>
> $T_d = 14.26$ Nm ; $P_d = 1.79$ kW

(Source: Adapted from exercise 5.8 in [Guru & Hızıroğlu, 2003])

Exercise 3

Consider a DC generator with $2p = 6$ poles, a simple lap armature winding, i.e. $2a = 6$ parallel paths, and $z = 624$ active conductors uniformly distributed around the armature periphery.

The cross-sectional area of each armature conductor is $s = 8$ mm^2, and carries a current density of $J = 3$ A.mm^{-2}.

The machine has a pole-arc-to-pole pitch ratio, $b/\tau = 2/3$, and the armature has a diameter of $D = 220$ mm.

a) Determine the magnetomotive force (MMF) associated with the armature reaction, $\mathcal{F}_i$, along the geometric neutral axis;

b) Assuming that the machine is fitted with interpoles only, determine the number of turns to be placed on the interpoles;

c) Suppose the the machine is to have its $\mathcal{F}_i$ fully compensated:

   1) Calculate the number of turns to be placed in a compensating winding, assuming that 6 slots were opened in each main pole. Calculate the number of conductors to be placed in each slot;

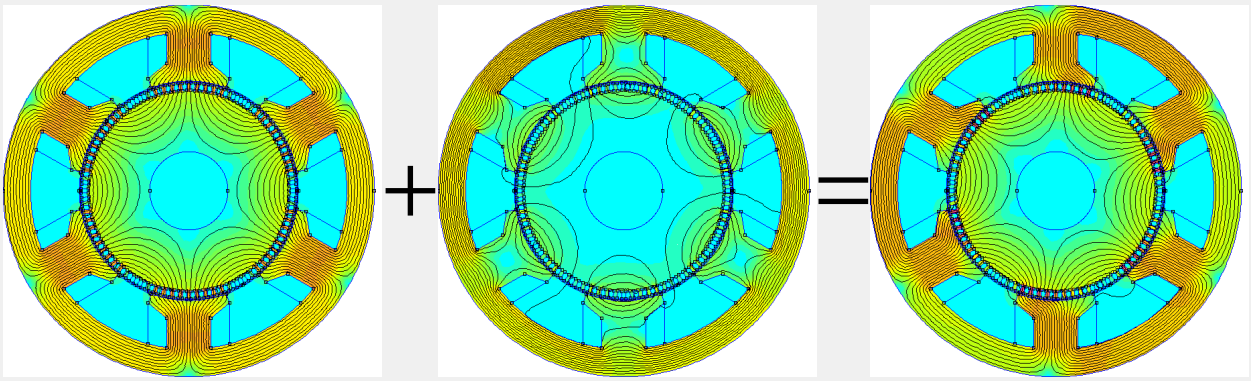
   2) Size the number of turns to be placed on the interpoles after the compensating windings have been installed.

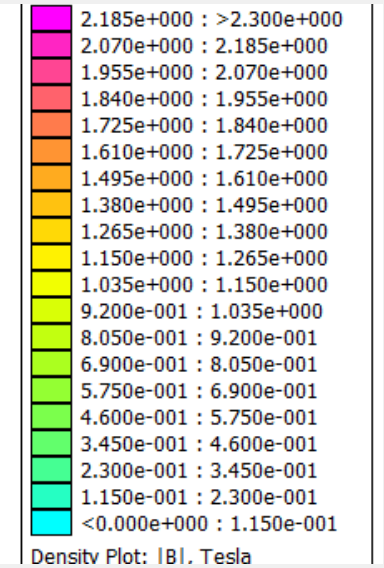
Worked solution:

Two solution approaches were used in this exercise:

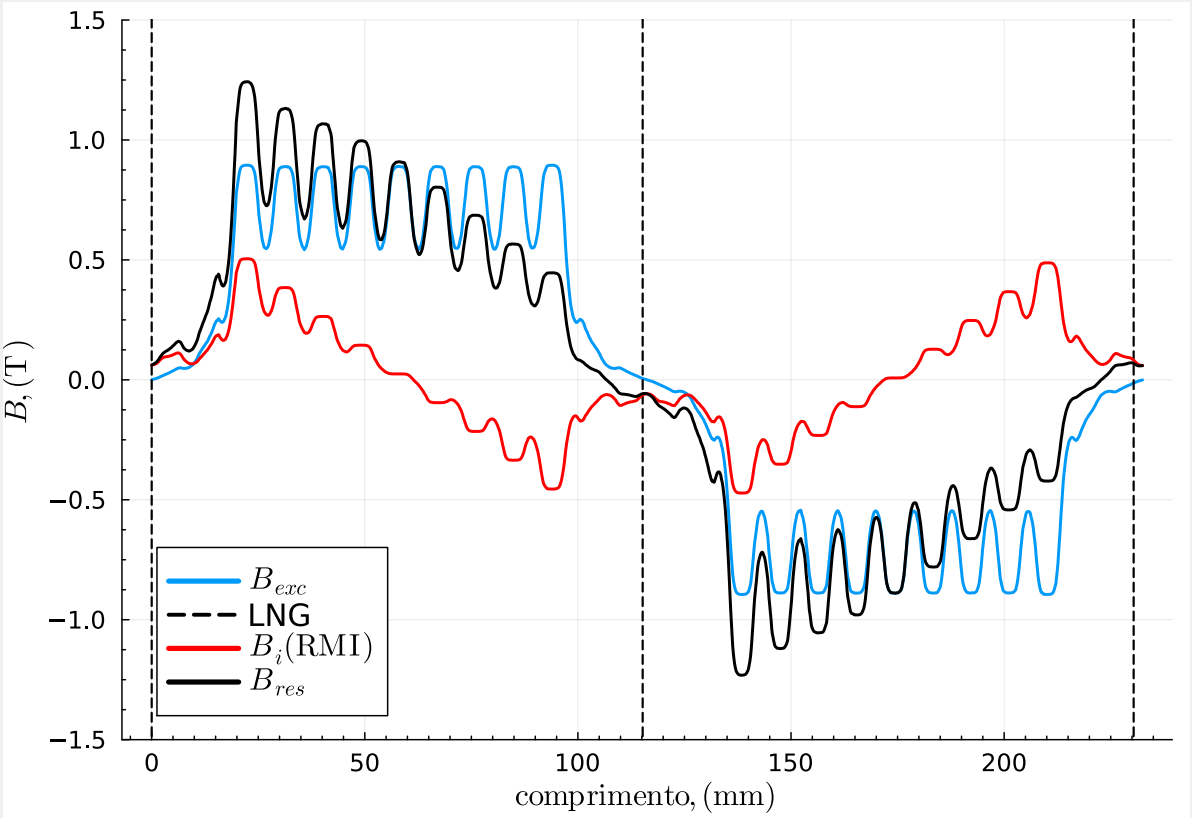
- **numerical (computational) solution**: using the exercise data, a CAD (*Computer-Aided Design*) and CAE (*Computer-Aided Engineering*) model was built, applying the finite element method for the electromagnetic calculation. The *software* used was FEMM - Finite Element Method Magnetics. The computational solutions make it possible to visualise and quantify the magnetic flux lines, helping to understand the armature magnetic reaction and its effects on the machine's operation;
- **Analytical solution**: this is the main objective of this exercise, namely to compensate the Armature Magnetic Reaction (AMR), by calculating the magnetomotive forces involved.

field winding + armature field = resultant field

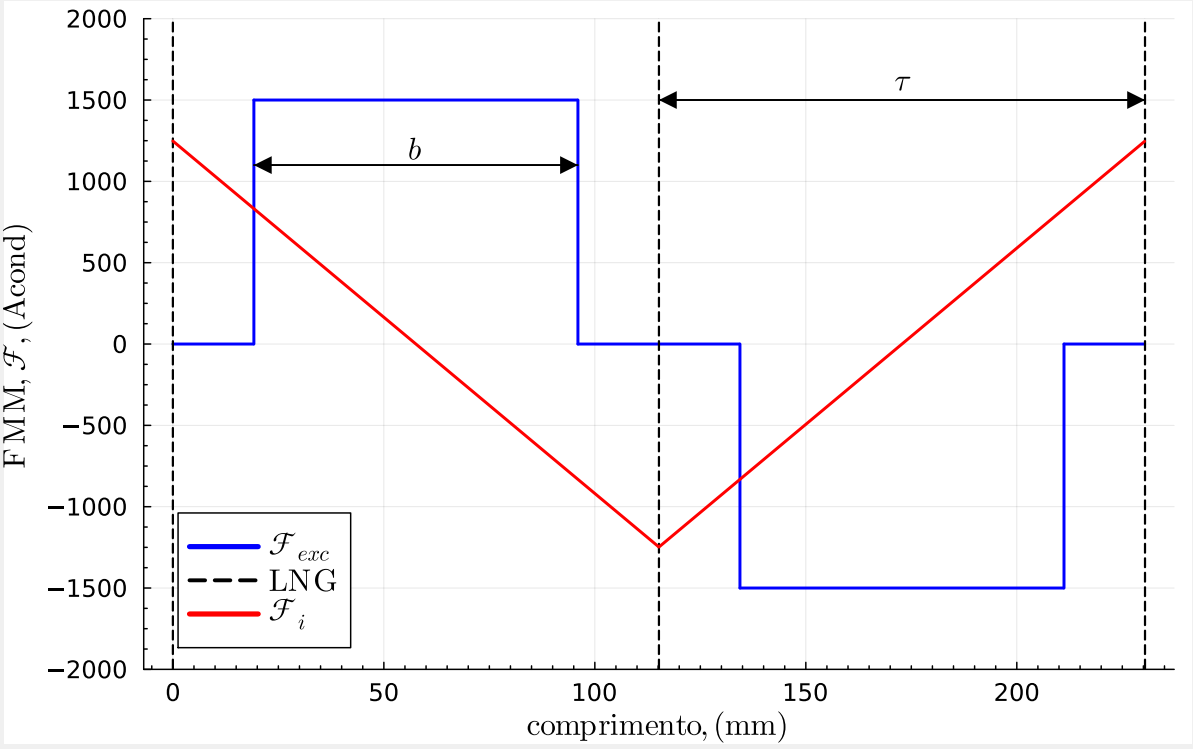




field winding + armature field = resultant field



MMF of the field winding and the armature reaction



a)

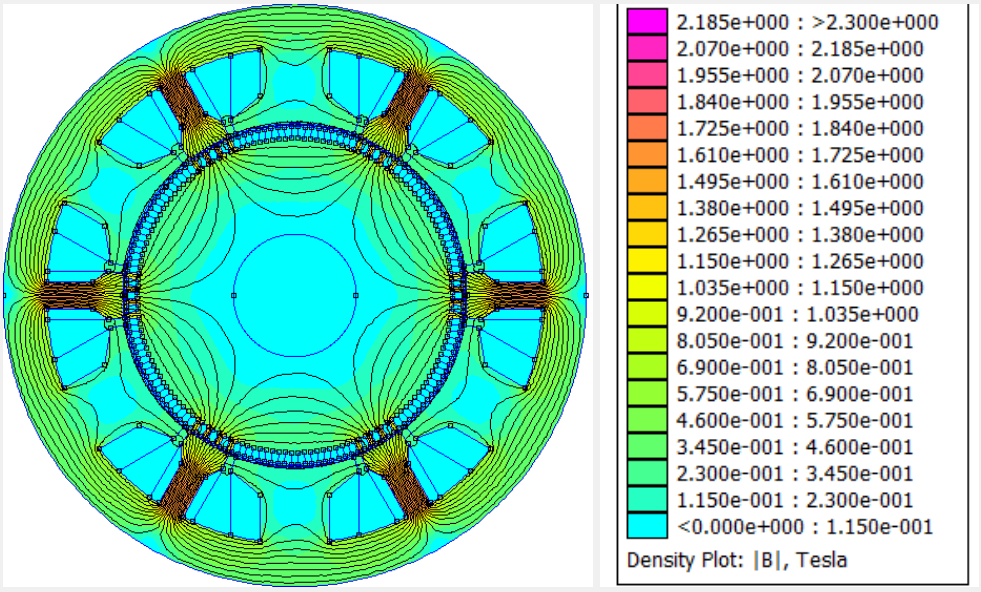
$I_a = J\ s = 3 \times 8 = 24\ \ \text{A}$

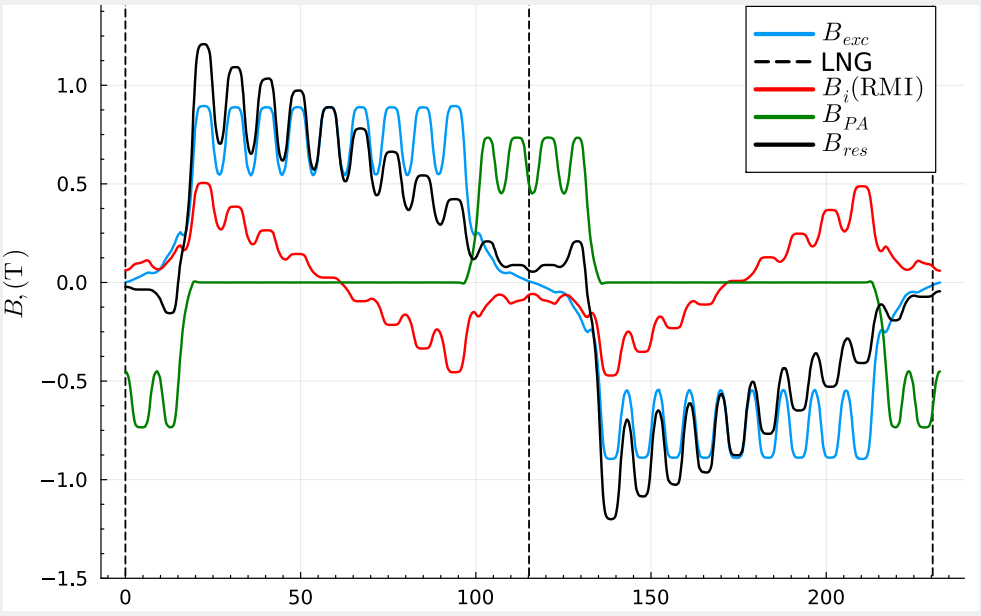
$I = I_a\ 2a = 24 \times 2 \times 3 = 144\ \ \text{A}$

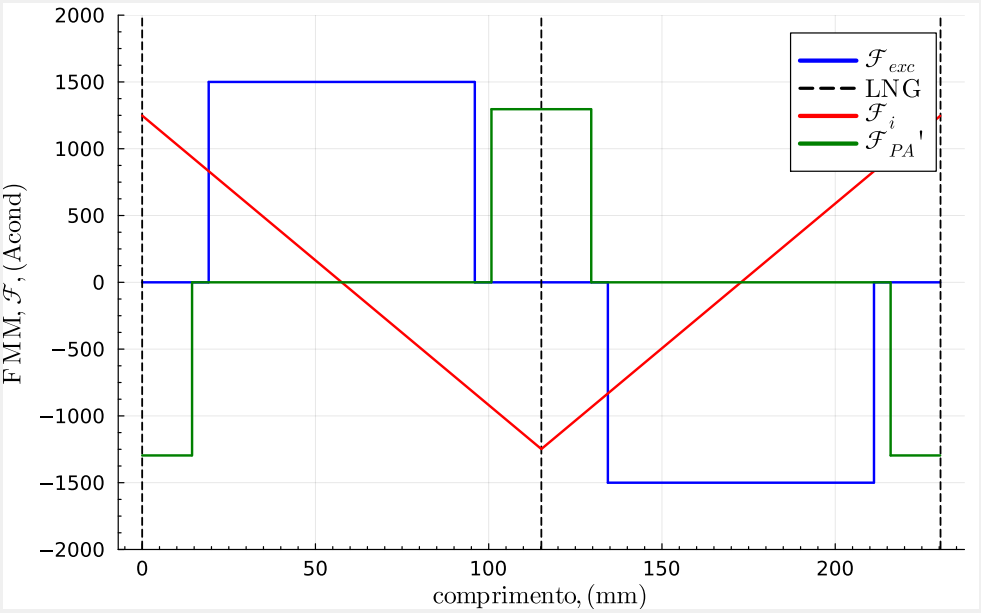
$\mathcal{F}_i = A\ \frac{\tau}{2}$ with: $A = \frac{z\ I}{4\ a\ p}\ \frac{1}{\tau}$ or $\mathcal{F}_i = \frac{z\ I}{8\ a\ p}$, using this latter expression:

$\mathcal{F}_i = \frac{z\ I}{8\ a\ p} = \frac{624 \times 144}{8 \times 3 \times 3} = 1248\ \text{A.cond}$

b) DC machine with **interpoles**:







For a machine with interpoles:

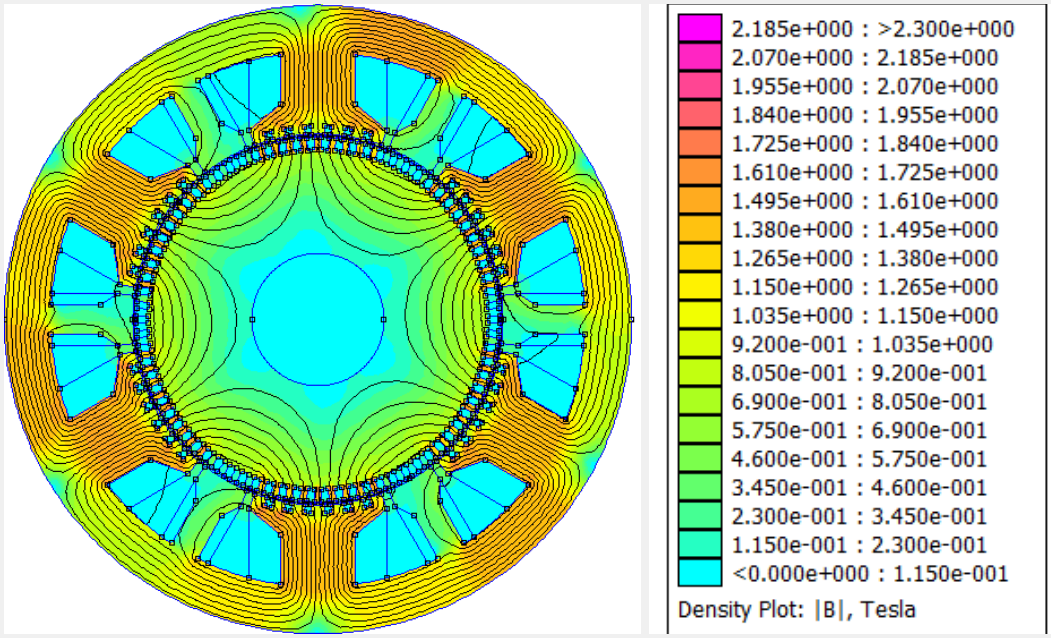
$\mathcal{F}_{\mathrm{PA}} = \mathcal{F}_{\mathrm{i}}$ since: $\mathcal{F}_{\mathrm{PA}} = N_{\mathrm{PA}}\ I$, we obtain: $N_{\mathrm{PA}} = \dfrac{\mathcal{F}_{\mathrm{i}}}{I} = \dfrac{1248}{144} = 8.67$ turns

To facilitate commutation, the next integer above is used, i.e. 9 turns.

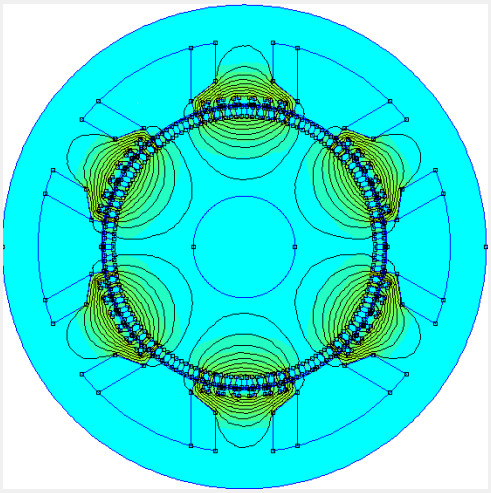
Thus, a corrected MMF is obtained for the interpoles, $\mathcal{F}'_{\mathrm{PA}}$:

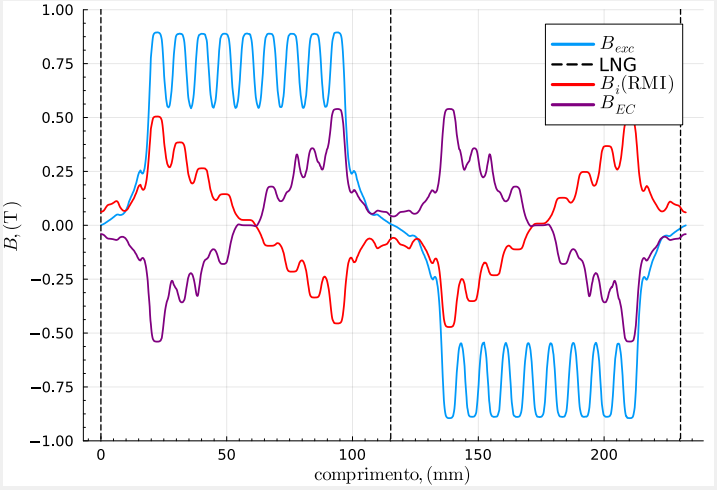
$\mathcal{F}'_{\mathrm{PA}} = 9 \times 144 = 1296$ A.cond

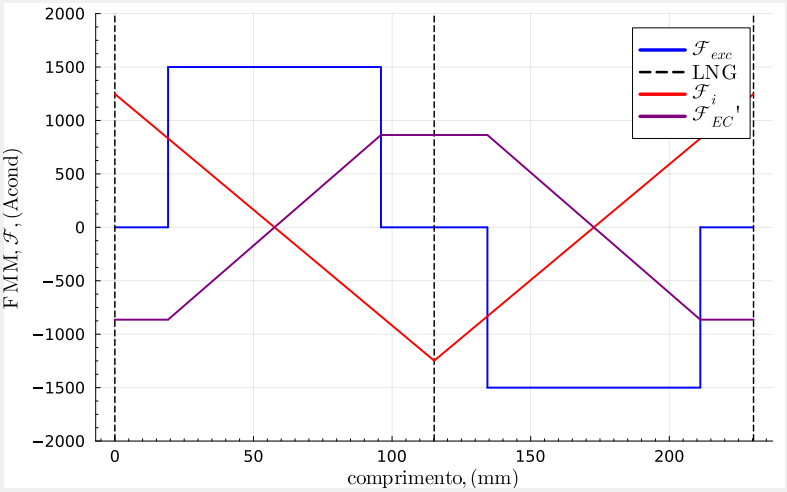
c) Machine with compensated AMR (**interpoles and compensating windings**)



c-1) The **compensating windings** are sized first:







$\mathcal{F}_{\mathrm{EC}} = A\,\dfrac{b}{2}$ or $\mathcal{F}_{\mathrm{EC}} = \mathcal{F}_{\mathrm{i}}\,\dfrac{b}{\tau}$, using this latter expression since $\dfrac{b}{\tau}$ is known:

$$\mathcal{F}_{\mathrm{EC}} = \mathcal{F}_{\mathrm{i}}\,\frac{b}{\tau} = 1248 \times \frac{2}{3} = 832 \text{ A.cond}$$

On the other hand, since $\mathcal{F}_{\mathrm{EC}} = N_{\mathrm{EC}}\ I$, the number of turns of the compensating winding is obtained as:

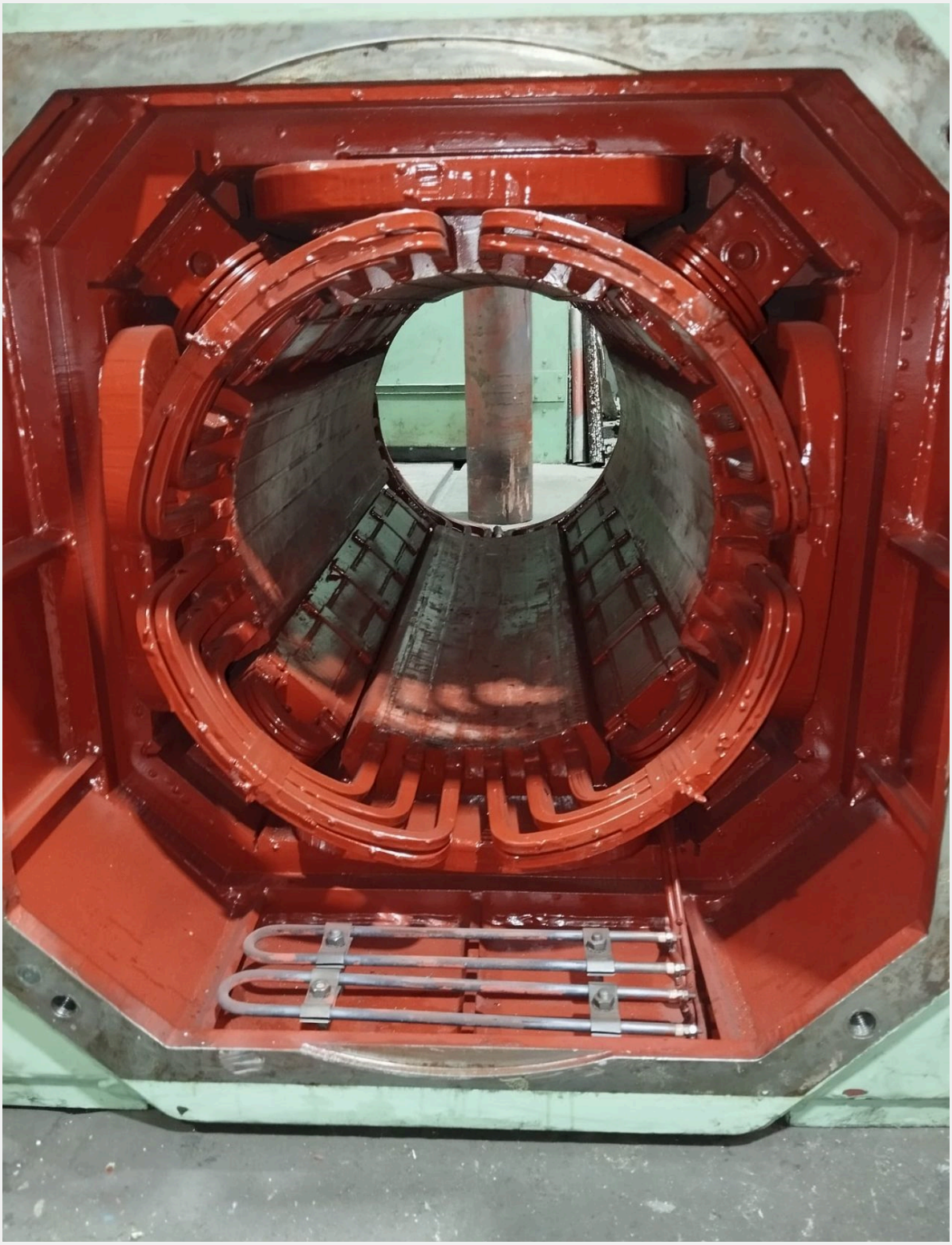
Thus, $\quad N_{\mathrm{EC}} = \dfrac{\mathcal{F}_{\mathrm{EC}}}{I} = \dfrac{832}{144} = 5.78 \text{ turns}$

Considering 6 slots per main pole, where the turns from 3 slots connect to the following pole and those from the other 3 slots connect to the preceding pole, $N'_{\mathrm{EC}} = 6$ turns are obtained, placing 2 turns in each slot.

Thus, a corrected MMF is obtained for the compensating windings, $\mathcal{F}'_{\mathrm{EC}}$:

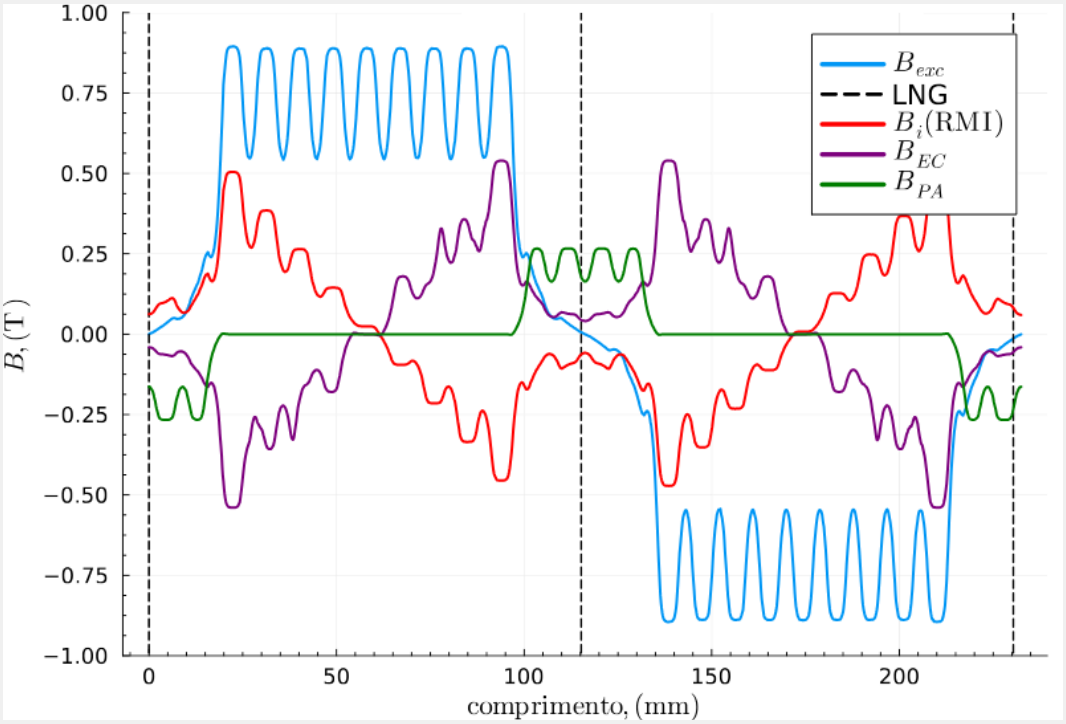
$$\mathcal{F}'_{\mathrm{EC}} = 6 \times 144 = 864 \text{ A.cond}$$

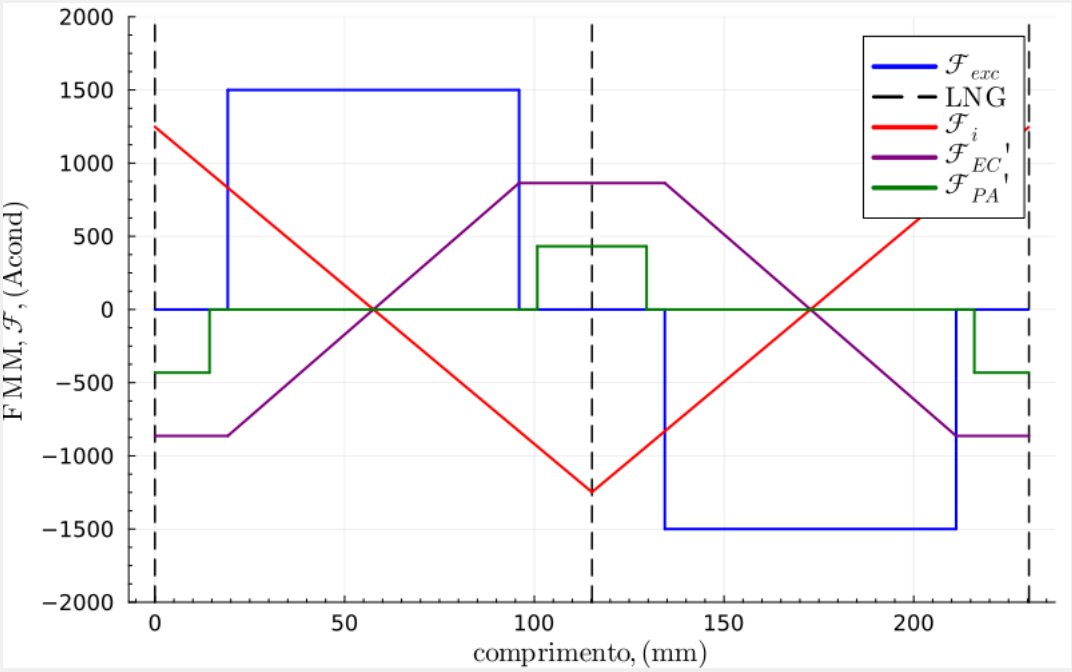
Example photograph of a machine with compensating windings (and interpoles) with 6 slots in the pole-shoe head and 2 turns per slot:



Photograph from refurbishment work on a DC machine.
Image of the stator with interpoles and compensating windings (source: Lusonave).

c-2) and then the **interpoles**:





$$\mathcal{F}_{\mathrm{PA}} = \mathcal{F}_{\mathrm{i}} - \mathcal{F}'_{\mathrm{EC}} = 1248 - 864 = 384 \text{ A.cond}$$

$$N_{\mathrm{PA}} = \frac{\mathcal{F}_{\mathrm{PA}}}{I} = \frac{384}{144} = 2.67 \text{ cond}$$

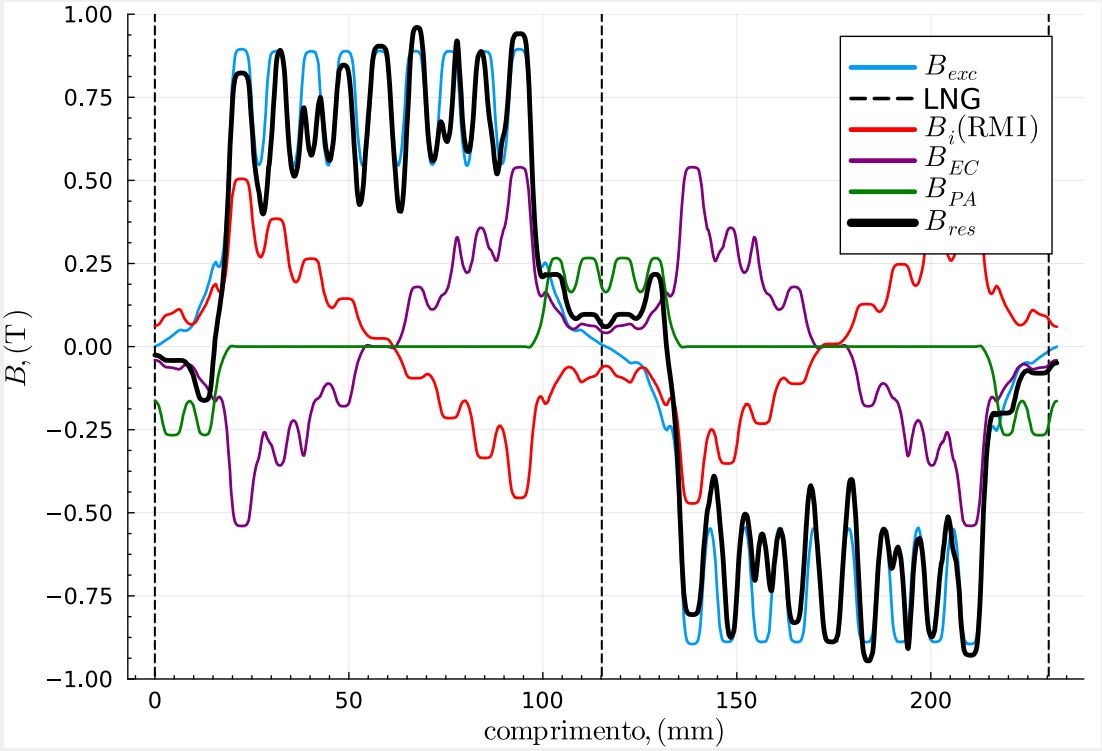
To facilitate commutation, the next integer above is used, i.e. 3 turns.

Thus, a corrected MMF is obtained for the interpoles, $\mathcal{F}'_{\mathrm{PA}}$:

$\mathcal{F}'_{\mathrm{PA}} = 3 \times 144 = 432$ A.cond

Conclusion: after fitting the compensating windings and the interpoles, a resultant MMF, $\mathcal{F}_R$, is obtained at the Geometric Neutral Lines (brush positions):

$\mathcal{F}_R = \mathcal{F}_i - \mathcal{F}'_{\mathrm{PEC}} - \mathcal{F}'_{\mathrm{PA}} = 1248 - 864 - 432 = -48$ A.cond



Exercise 4

An armature winding of a dynamo has $z = 640$ conductors, wound as a simple lap winding, for a rated current of 100 A and 2 pole pairs.

a) Size the number of turns of the compensating windings, N_{EC}, and of the interpoles, N_{PA}, for full compensation of the armature magnetic reaction. Consider that the machine's field poles cover 70 % of the total periphery, as seen from the rotor;

b) Resize N_{EC} and N_{PA} considering a simple wave armature winding.

Results:

a) $N_{\mathrm{EC}} = 14$ turns; $N_{\mathrm{PA}} = 6$ turns

b) $N_{\mathrm{EC}} = 28$ turns; $N_{\mathrm{PA}} = 12$ turns

Exercise 5

A separately-excited DC generator $[220\ \mathrm{V},\ 12\ \mathrm{A},\ 1500\ \mathrm{rpm}]$ was tested at no-load and under load, at rated speed, yielding the following characteristics:

E_0 [V]	20	180	238	270	284	300
i [A]	0.00	0.25	0.50	0.75	1.00	1.50

U [V]	278	260	242	216	186	156	128	96	62
I [A]	0	5	10	15	20	25	30	35	40

a) Determine the total internal voltage drop of this generator;

b) What is the critical resistance of a shunt-excited generator? What is its significance? How is it determined (approximately) in practice?

c) What is the field winding resistance, given that, when operating as a shunt generator, at rated speed, with no field rheostat, $U_0 = 294$ V;

d) Explain qualitatively the influence that varying the field rheostat has on the external characteristic of the shunt generator. Justify briefly;

e) Under the excitation conditions of part c), what should be done to obtain a no-load voltage of 336 V?

Worked solution to Exercise 5: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 6

A shunt-excited generator has the following no-load (magnetisation) characteristic, at 1500 rpm:

i[A]	0	0.1	0.2	0.3	0.4	0.6	0.8	1.0
E[V]	20	120	200	235	250	270	285	300

$R_a = 2.0\ \Omega$

$R_d = 300\ \Omega$

Consider that the generator is fitted with interpoles and compensating windings.

a) With the generator driven at 1500 rpm, determine the value of the field rheostat to obtain the operating point $(35\ \text{A}; 180\ \text{V})$;

b) Determine the generator drive speed required for the machine operate under the same conditions as in part a), without a field rheostat;

c) Under the conditions of part a) (1500 rpm and field rheostat), determine the equivalence ratio (N_s/N_d), to obtain the operating point $(35\ \text{A};\ 200\ \text{V})$. Consider that the series field winding has a resistance of $0.5\ \Omega$.

Results:

a) $150\ \Omega$; **b)** 1389 rpm; **c)** $0.39/35$

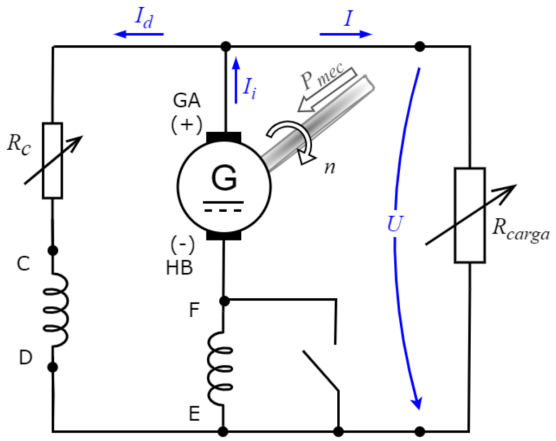
Exercise 7

A long-shunt compound generator, with cumulative (additive) series field flux, has the following characteristics:

$P = 8.75$ kW	$U = 250$ V	$R_d = 223\ \Omega$	$R_a = 1.3\ \Omega$	$R_s = 0.1\ \Omega$
$n = 1500$ rpm	$n_{\text{mag}} = 1500$ rpm	$N_d = 2000$ turns	$N_s = 50$ turns	$E_0 = \text{f}(I_{\text{exc}})$:

I_{exc} [A]	0.0	0.2	0.4	0.6	0.8	0.9	1.0	1.1	1.3	1.5	1.7
E_0 [V]	15	100	170	220	250	260	270	280	290	300	310

Also consider a switch connected in parallel with the series field winding, as show in the following electric diagram:



a) Calculate the no-load voltage with no field rheostat. Explain whether the state of the switch influences the generator no-load voltage;

b) With the switch open, calculate the voltage drop due to the armature reaction, knowing that flat (level) regulation is obtained under rated conditions;

c) With the switch closed, explain qualitatively the generator external characteristic. What is the variation of the operating point on the external characteristic, for a given load resistance, in the following situations:

1) an increase in drive speed;

2) a decrease in the shunt-field rheostat.

Worked solution to Exercise 7: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 8

Consider a series generator with 5 turns/pole, whose magnetisation characteristic passes through the following points, at 1200 rpm:

$\mathcal{F}$ [Ae]	0	200	600	1000
E_0 [V]	5	120	160	170

It is known that the rotational losses are 1200 W, $R_a = 0.2\ \Omega$ and $R_s = 0.05\ \Omega$.

a) Determine the voltage drop due to the armature reaction, when supplying a $1\ \Omega$ load, at 120 A;

b) Consider that the armature reaction results in a flux loss of $5\ \%$. Will it be possible to obtain a voltage of 150 V, at the same current? Justify;

c) An attempt was made to operate this series dynamo on a $3\ \Omega$ load, but this was not possible. Comment on the situation;

d) Determine the current value corresponding to maximum efficiency.

Results:

a) 10 V ; **d)** 69.3 A

Exercise 9

The external characteristics of two shunt-excited dynamos, rated at 220 V, 110 kW, and connected in parallel, are known:

I [A]	0	200	400	500	700	900
U_1 [V]	229.5	226.5	222.5	220.0	213.0	205.5
U_2 [V]	224.0	223.0	221.0	220.0	217.5	214.0

a) How would the two machines share a current of 1500 A? What is the voltage?

b) What happens for small loads, close to zero? $0 < I < 250$ A

c) Complete:

"Under overload, the machine with _____ regulation supplies _____ current".

Worked solution to Exercise 9: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 10

Two identical shunt-excited dynamos have the following magnetisation characteristic at 1500 rpm:

i [A]	0.0	0.2	0.6	1.0	1.4
E [V]	20	200	220	240	260

The armature resistance is $0.4\ \Omega$ and the field winding resistance is $150\ \Omega$; each also has a field rheostat connected, with a maximum value of $300\ \Omega$.

The two machines are connected in parallel, supplying a load $(100\ \text{A},\ 200\ \text{V})$, with **dynamo A** running at 1500 rpm.

a) Initially **dynamo B** is unloaded. Determine the field rheostat of **dynamo A**;

b) Determine the short-circuit current of **dynamo A**;

c) **Dynamo A** supplies 75 A of the total load. How should one proceed so that **dynamo B** supplies the rest? Show the calculations.

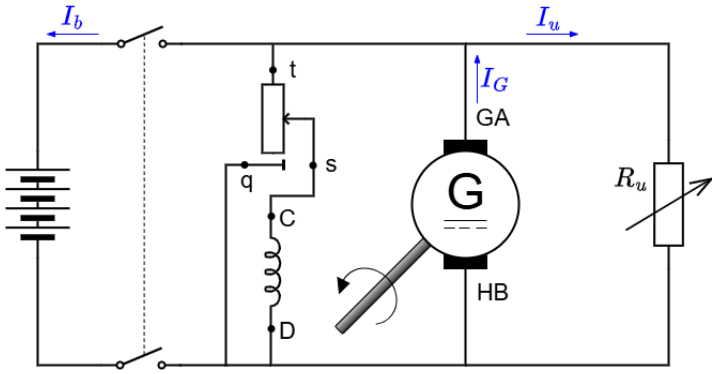
Results:

a) $R_f^{\mathrm{A}} = 50\ \Omega$; **b)** 50 A

c) Adjusting only on the field rheostat results in $R_f^{\mathrm{B}} > R_f^{\mathrm{máx}}$, so the speed must also be adjusted. For example, considering $R_f^{\mathrm{B}} = 50\ \Omega$, 1312.5 rpm is obtained.

Exercise 11

Consider a shunt generator driven at constant speed, supplying a load resistance $R_u = 7\ \Omega$. A lead-acid battery is occasionally connected to this circuit for charging, according to the following electrical circuit:



External characteristic of the generator:

I_G [A]	0.0	10	20	30	40	50	60
U_G [V]	312	300	284	262	234	200	160

The battery consists of 100 cells in series and is charged whenever its SOC (State Of Charge) reaches $25\ \%$ (approximately 1.95 V/cell). The internal resistance of each battery cell is 8 mΩ.

Determine the operating voltage of the circuit when the battery is connected (generator + load R_u), the generator current, I_G, the current in the load, I_u, and the initial battery charging current, I_b.

Worked solution:

The battery open-circuit voltage, E_b, for SOC$= 25\ \%$, is given by:

$$E_b = 100 \times 1.95 = 195\ \mathrm{V}$$

The internal resistance of the battery, R_b, with its cells in series:

$$R_b = 100 \times 8 \times 10^{-3} = 0.8\ \Omega$$

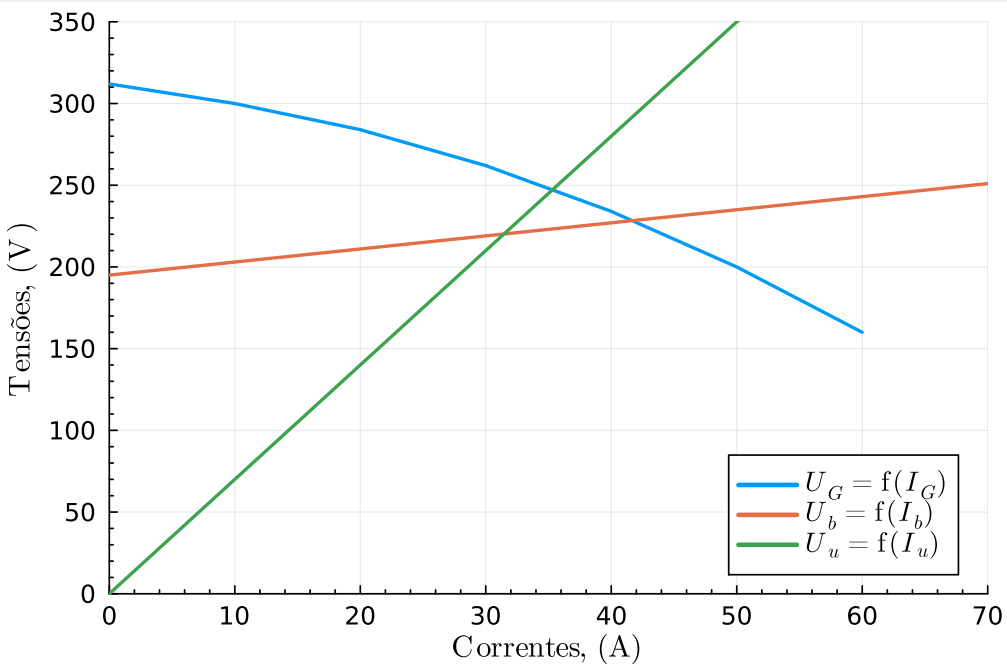
The battery charging equation, $U_b = \mathrm{f}\,(I_b)$, is given by:

$$U_b = E_b + R_b\, I_b \ \Leftrightarrow\ U_b = 195 + 0.8\, I_b$$

The following graph shows:

- the generator external characteristic, $U_G = \mathrm{f}(I_G)$
- the load-resistor line, $U_u = \mathrm{f}(I_u)$
- the battery charging line, $U_b = \mathrm{f}(I_b)$

The generator loads (the load resistor and the battery) must be combined in order to determine the operating point of the generator and of the loads.



The combined load must reflect the operation of the circuit, that is:

- the voltages across each branch are equal: $U = U_G = U_u = U_b$
- the generator supplies the load resistor and the battery: $I = I_G \approx I_u + I_b$

Note: The generator field current, I_d, is much smaller than the generator armature current and the load currents, so it is neglected in the graph:

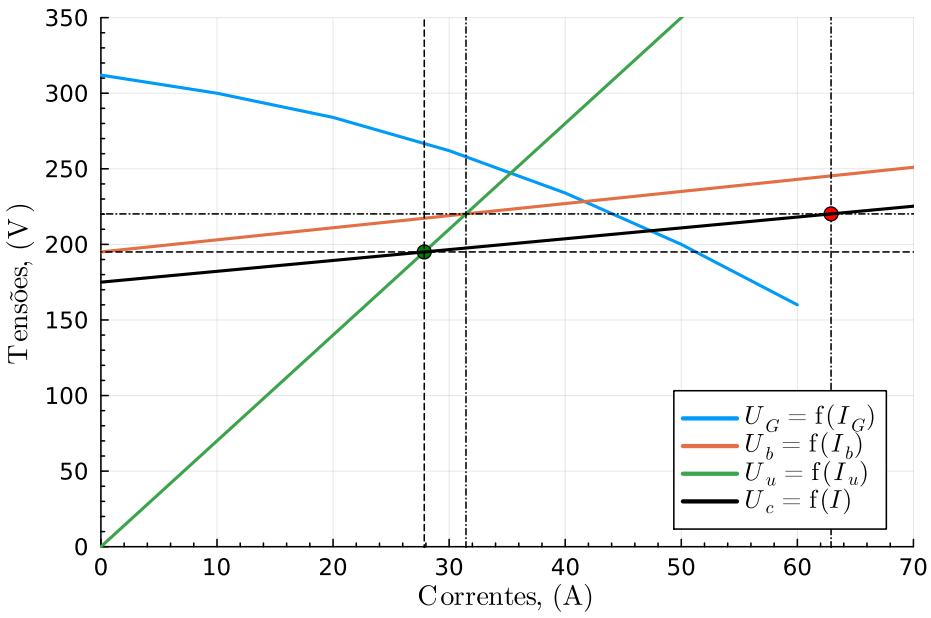
$$I_d \ll \{I_b, I_u, I_G\}$$

The following graph illustrates the graphical determination of the combined load characteristic, $U_c = \mathrm{f}\ \ (I)$, with $I = I_b + I_u$, corresponding to the combined loads. Since the sum of two straight lines is another straight line, only 2 voltage levels need to be considered to obtain 2 points on the combined load characteristic. Thus:

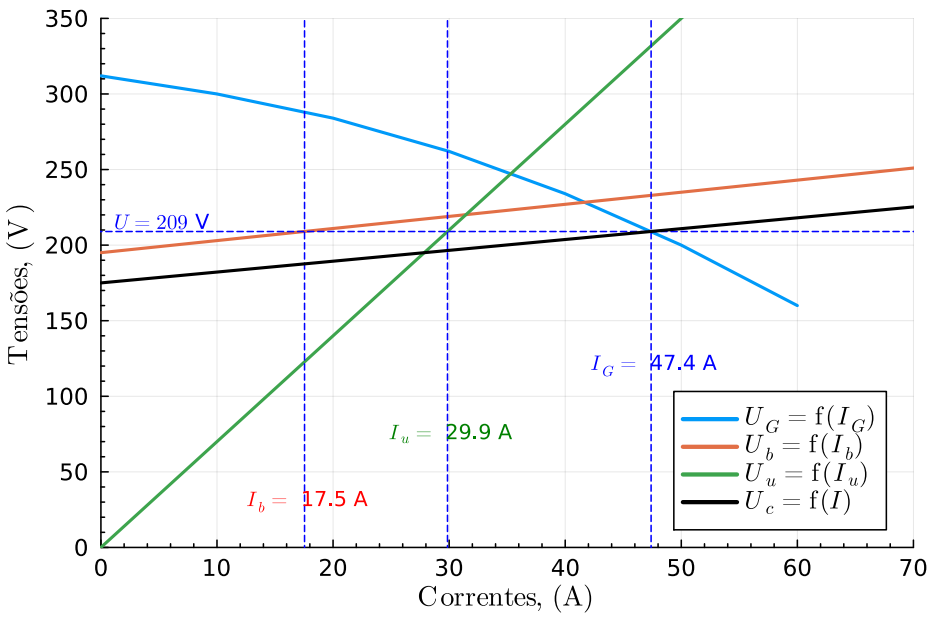
- the green point was obtained by choosing the voltage level corresponding to the battery being disconnected, i.e. E_b. In this situation, $I_b = 0$ and therefore $I = I_u$
- the red point was obtained by choosing the voltage level at which the load-resistor and battery lines intersect, i.e. $I_u = I_b$. At this voltage level, the red point is placed at the abscissa corresponding to twice the current.

Joining the points (green and red) gives the combined load characteristic:

$U_c = \mathrm{f}\ \ (I)$



The intersection of the combined load characteristic with the external characteristic of the DC generator defines the operating point of the circuit: $(I_G; U) = (47.4\ \mathrm{A}; 209\ \mathrm{V})$.



On the other hand, from the operating voltage, $U = 209$ V, the load lines of the load resistance and of the battery are consulted to obtain the respective current values, I_u and I_b, supplied by the generator:

$$I_u = 29.9 \text{ A} \qquad ; \qquad I_b = 17.5 \text{ A}$$

As an alternative to the graphical determination of the combined load characteristic, since it corresponds to the sum of two straight lines, it can be determined analytically. Given that:

$$U_c = U_b = U_u \quad \text{and that} \quad I = I_b + I_u \quad \text{then:}$$

$$I = \frac{U_c - E_b}{R_b} + \frac{U_c}{R_u}$$

Rearranging the previous equation the combined load line (load resistance + battery), $U_c = \text{f}\ (I)$:

$$U_c = \frac{R_u}{R_u + R_b} E_b + \frac{R_u R_b}{R_u + R_b} I$$

Exercise 12

A series motor with an armature resistance of $0.8\ \Omega$ and a field winding resistance of $0.4\ \Omega$ was run as a dynamo:

- Separately excited at 50 A and driven at 1500 rpm, it produced 230 V at the open-circuit terminals;
- With series excitation, delivering 50 A at the same speed, it produced 150 V.

Knowing that the mechanical and magnetic losses, $p_{(\text{mec}+\text{Fe})}$, are 400 W, calculate:

a) The speed when it draws 50 A as a series motor from a 220 V supply;

b) The mechanical torque it delivers to the load under these conditions;

c) Sketch qualitatively the torque and speed characteristics of this motor. Explain the effect of introducing a field rheostat on these characteristics.

Results:

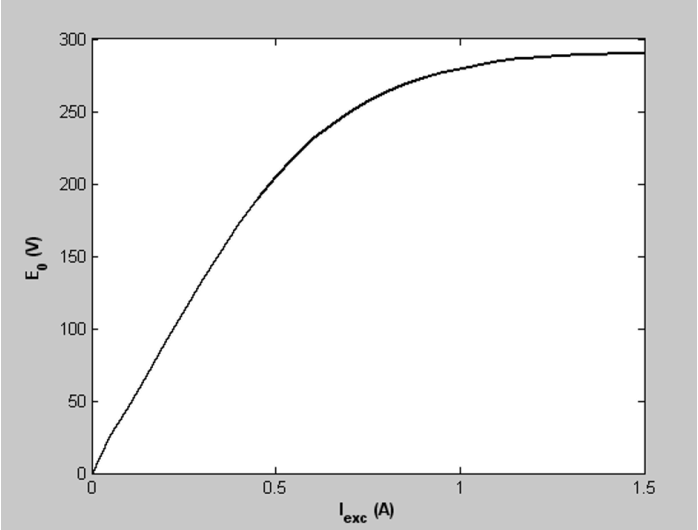
a) 1174 rpm ; **b)** 61.8 Nm

Exercise 13

Consider a DC motor with the following nameplate ratings: 17 kW, 250 V, 1200 rpm, $\eta = 85\ \%$.

The following parameters are also known:

$E_0 = f(I_{\text{exc}})$, $n = 1200$ rpm
$R_a = 0.6\ \Omega$
$R_s = 0.1\ \Omega$, 30 turns
$R_d = 200\ \Omega$, 3000 turns



a) With the motor shunt-excited, determine the value of the field rheostat under rated conditions (U_n, I_n, n_n);

b) Using the field rheostat calculated in the previous part, determine the speed, torque and mechanical characteristics of this motor (shunt excitation);

c) Repeat for cumulative and differential compound, long-shunt excitation. Plot the characteristics on the same graphs for comparison;

d) Determine the characteristic curves with the shunt circuit disconnected (series motor). Plot the characteristics on the same graphs for comparison;

e) Consider the motor with separate excitation, $U_{\text{exc}} = 240$ V, with the field rheostat calculated in part a). Explain how the speed characteristic changes in the following situations:

   1) an increase in armature voltage;

   2) a decrease in the field rheostat;

   3) an increase in the additional resistance.

Worked solution to Exercise 13: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 14

A shunt-excited motor has the following characteristics:

$$U_n = 220\ \text{V} \qquad I_n = 42\ \text{A} \qquad n_n = 1500\ \text{rpm} \qquad R_a = 0.34\ \Omega$$

a) In the no-load test the following values were obtained:

$$U = 220\ \text{V} \qquad n = 1600\ \text{rpm} \qquad I_a = 3\ \text{A} \qquad I = 4.6\ \text{A}$$

Calculate the loss torque;

b) Obtain the speed equation, $n = \text{f}\ (I_a)$;

c) Size a stepped starting rheostat such that: $40\ \text{A} \leq I_{\text{start}} \leq 65\ \text{A}$.

Results:

a) 3.92 Nm **b)** $n(\text{rpm}) = \dfrac{\text{-}100}{37.4}(I_a - 3) + 1600$

c) Resistances between the 6 contacts (Ω): 1.30; 0.80; 0.49; 0.30; 0.15. Rheostat: $3.04\ \Omega$.

Exercise 15

A series motor rated at 12 kW, 250 V, 1400 rpm, with an efficiency of $80\,\%$ and a maximum speed of 2400 rpm, has the following magnetic characteristic obtained at 1500 rpm:

I_{exc} (A)	10	20	30	40	50	60	70	80
E_0 (V)	80	140	190	225	250	270	285	295

Knowing that the armature resistance is $0.35\ \Omega$ and the field resistance is $0.1\ \Omega$, calculate:

a) The mechanical and iron losses, $p_{(\text{mec+Fe})}$;

b) The minimum current the motor can draw;

c) The voltage drop due to armature reaction at full load;

d) The motor power corresponding to maximum efficiency;

e) Explain qualitatively the effect of the field rheostat on the speed characteristic of the series motor;

f) Explain qualitatively the effect of the field rheostat on the torque characteristic of the series motor.

Worked solution to Exercise 15: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 16

A series motor supplied at 250 V has an armature resistance of 0.2 Ω and a field winding resistance of 0.15 Ω. This field winding has a field rheostat of 0.1 Ω fitted with a switch.

a) Assume the switch is open. Under these conditions, the motor delivers a shaft torque of 25 Nm, running at 800 rpm and drawing a current of 10 A.
Calculate the efficiency, η (%), the electromagnetic torque, T_d, and the mechanical and magnetic losses, $p_{(\mathrm{mec+Fe})}$;

b) Assume the switch is closed. What are the new values of current drawn and speed when the electromagnetic torque doubles? State the assumptions made;

c) Sketch qualitatively the shaft torque characteristic of this motor. Explain what happens to it when the supply voltage is varied. Justify.

Results:

a) 83.8 % ; 29.4 Nm ; 370.6 W **b)** 22.4 A ; 888 rpm

Exercise 17

A shunt motor with the following nameplate ratings: 45 kW, 500 V, 100 A, 1600 rpm, has an armature resistance of 0.33 Ω and will use reverse current braking (plugging), from its rated operating condition.

Knowing that the initial braking current must be limited to 150 A, calculate:

a) The useful (shaft) torque, T_u, and the back electromotive force, E', at full load;

b) The value of the braking resistance;

c) Analyse qualitatively the torque developed at the instant the motor stops. What happens from that instant onwards?

d) Consider dynamic (rheostatic) braking. Calculate the initial braking current using the braking resistance found above, starting from the rated operating condition.

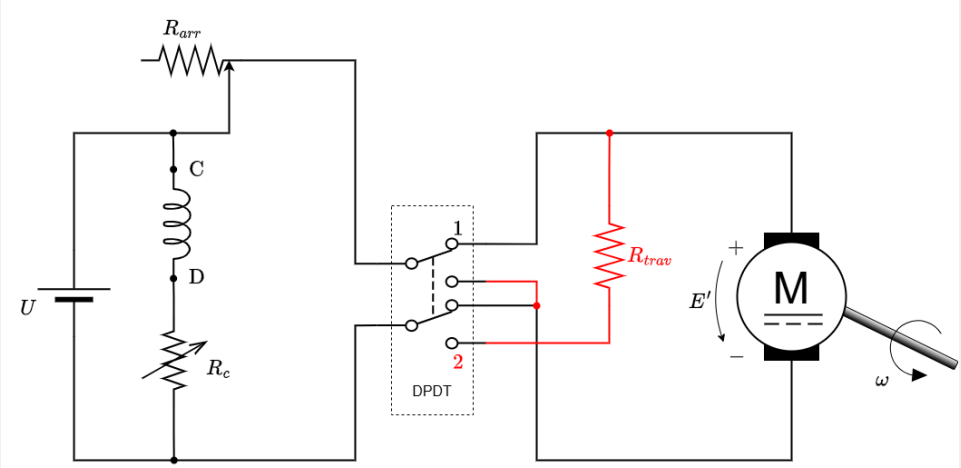
Worked solution:

a) The shaft torque is obtained from the nameplate ratings:

$$T_u = \frac{P_u}{\frac{2\,\pi\,n_n}{60}} = \frac{45\,000}{\frac{2\,\pi\,1600}{60}} = 268.6 \text{ Nm}$$

Calculation of E': $\quad E' = U - R_a\,I_n = 500 - 0.33 \times 100 = 467 \text{ V}$

b) The following figure shows the circuit diagram of the shunt motor fitted with a DPDT (Double Pole, Double Throw) switch for the **reverse current braking (plugging)** manoeuvre:



When the DPDT switches to perform reverse current braking (position 2), the negative armature terminal is connected to the positive terminal of the DC supply, while the positive motor terminal is connected to the negative terminal of the supply through a braking resistance, R_{brk}.

Thus, going round the power loop, the following equation is obtained:

$$-U - E' + (R_a + R_{\text{brk}})\,I_{\text{brk}} = 0 \quad \Rightarrow \quad I_{\text{brk}} = \frac{U + E'}{R_a + R_{\text{brk}}}$$

Note that the braking current, I_{brk}, flows in the opposite direction to the motor current with the DPDT in position 1. Therefore, the torque developed during braking opposes the rotor motion, forcing the speed, ω, to decrease.

As the speed, ω, decreases, the back electromotive force, E', decreases and consequently the current and the torque during the DC motor braking operation also decrease.

Rearranging the previous expression to obtain the braking resistance, R_{brk}, gives:

$$R_{\text{brk}} = \frac{U + E'}{I_{\text{brk}}} - R_a = \frac{500 + 467}{150} - 0.33 = 6.12\ \Omega$$

c) Relating the developed torque, T_d, to the speed, ω, we obtain successively:

$$T_d = K\,\phi\,I_a$$

Particularising for the torque developed during braking, $T_{d_{\text{brk}}}$, considering I_{brk}:

$$T_{d_{\text{brk}}} = K\,\phi\left(\frac{U + E'}{R_a + R_{\text{brk}}}\right) \qquad \text{since:} \qquad E' = K\,\phi\,\omega$$

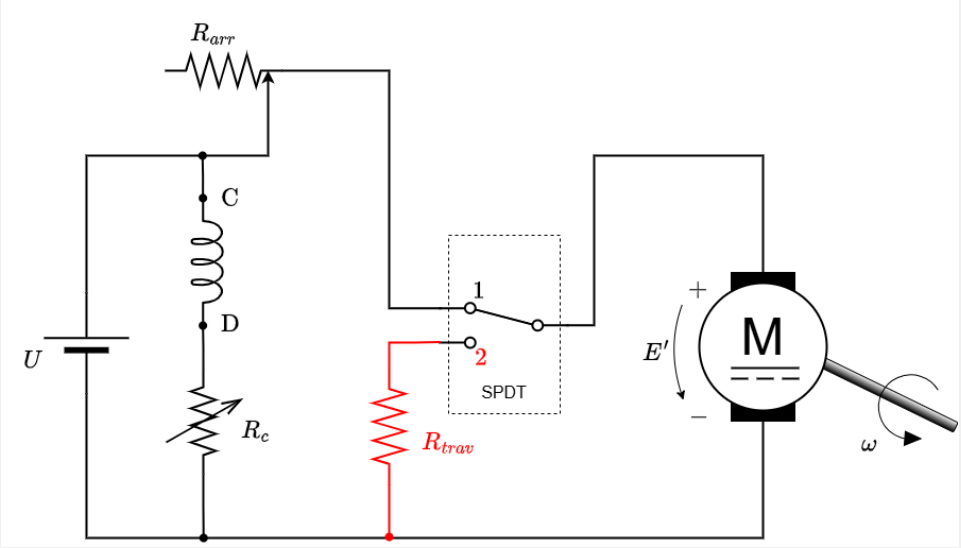
$$T_{d_{\text{brk}}} = \frac{K\,\phi\,U}{R_a + R_{\text{brk}}} + \frac{(K\,\phi)^2\,\omega}{R_a + R_{\text{brk}}} \quad \text{, when:} \quad \omega = 0 \;\Rightarrow\; T_{d_{\text{brk}}} = \frac{K\,\phi\,U}{R_a + R_{\text{brk}}}$$

Analysing the previous expression, the second term decreases during the braking manoeuvre until it becomes zero when: $\omega = 0$.

However, the first term remains constant. Therefore, immediately after the speed reaches zero, if no action is taken, the motor starts will start rotating in the opposite direction. An automatic control must therefore disconnect the motor from the DC supply as soon as its speed reaches zero.

d) Dynamic (or rheostatic) braking consists of disconnecting the motor armature circuit from the DC supply and connecting it across a braking resistance. The DC machine will now operate as a separately excited generator (constant flux) supplying the braking resistance. Since there is no prime mover driving the “generator”, the speed falls sharply within a few moments.

The electrical diagram in the following figure uses an SPDT (*Single Pole, Double Throw*) switch for the dynamic braking manoeuvre:

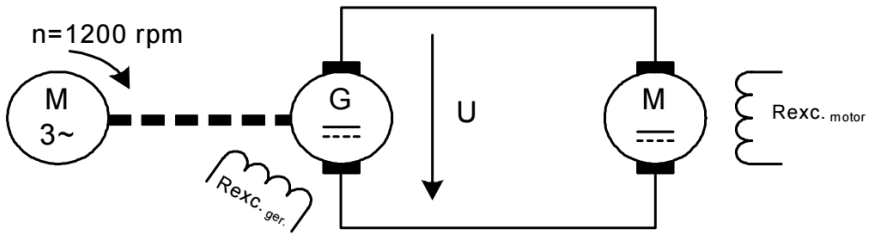


Analysing the flow of the braking current (with the SPDT switch in position 2), at the start of the braking operation, we obtain:

$$I_{\text{brk}} = \frac{E'}{R_a + R_{\text{brk}}} = \frac{467}{0.33 + 6.12} = 72.4\ \text{A}$$

Exercise 18

A separately excited DC motor, 10 kW, 250 V, with an efficiency of 88 %, is supplied by a Ward Leonard system, shown in the following schematic diagram of its operating principle:



The magnetisation characteristics of the generator and the motor, obtained at 1200 rpm, are known:

Generator:	i(A)	0.0	0.4	0.8	1.2	1.6	2.0	$R_a = 0.8\,\Omega$	$R_d = 120\,\Omega$
	E(V)	12	110	200	260	290	310	$U_{\text{exc}} = 240$ V	$\Delta E = 0$ V

Motor:	i(A)	0.0	0.4	0.8	1.2	1.6	$R_a = 1.6\,\Omega$	$R_d = 150\,\Omega$
	E(V)	10	105	190	240	265	$U_{\text{exc}} = 240$ V	$\Delta E = 0$ V

a) Calculate the motor speed and the electromagnetic torque at full load, with: $R_f^{\text{gen}} = 180\,\Omega$ and $R_f^{\text{mot}} = 50\,\Omega$;

b) Using the Ward Leonard system, adjust the motor speed to 1100 rpm at half load. Calculate the value of the field rheostat for this condition;

c) When the generator field current reaches its maximum value, how can the speed of the DC motor be increased? Justify;

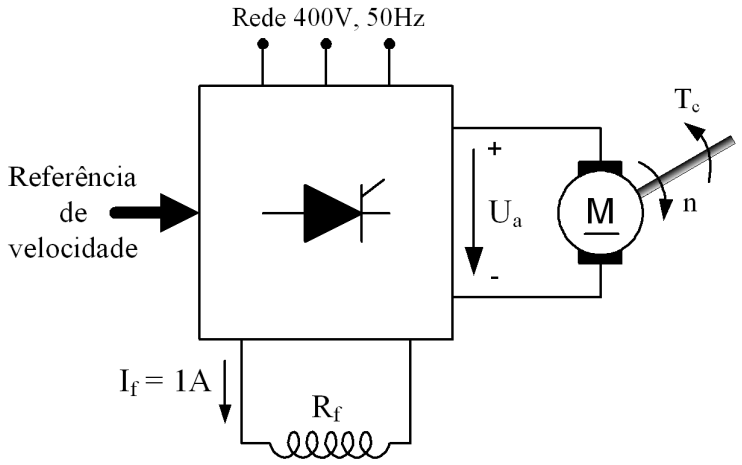
d) State the advantages of replacing the Ward Leonard system with an electronic variable-speed drive.

Results:

a) 455 rpm ; 87 Nm **b)** 53 Ω

Exercise 19

Consider a DC motor supplied from an electronic Ward-Leonard system, as shown in the following figure:



$P_n = 5$ kW

$T_n = 30$ Nm

$I_n = 40$ A

$R_a = 0.8\,\Omega$

The magnetisation characteristic of the machine, obtained at 2000 rpm, is also known:

I_f (A)	0	0.25	0.50	0.75	1.00	1.25	1.50
E_0 (V)	20	160	210	235	250	260	265

a) Determine the voltage applied to the motor armature under rated conditions;

b) Consider the motor operating under the following conditions: $U_a = 195$ V; $n = 1400$ rpm. Now suppose that, using the "Speed reference" command of the electronic variable-speed drive, the armature voltage is suddenly changed to 155 V. Calculate the machine current at the immediately after the change. What has happened? Justify.

Worked solution:

a) Using the rated values provided, the rated speed is obtained from:

$\omega = \dfrac{P_n}{T_n}$ combined with: $\omega = \dfrac{2\,\pi\,n}{60}$, we have:

$$n = \frac{P_n}{T_n}\,\frac{60}{2\,\pi} = \frac{5000}{30}\,\frac{60}{2\,\pi} = 1592 \text{ rpm}$$

On the other hand, for $I_f = 1$ A $\Rightarrow\ K\,\phi = \dfrac{250}{2000}$ V/rpm

Using the speed equation: $n = \dfrac{U_a - R_a\,I_n}{K\,\phi}$, with: $\Delta E = 0$, we obtain:

$$1592 = \frac{U_a - 0.8 \times 40}{250/2000} \ \Rightarrow\ U_a = 231 \text{ V}$$

b) For the specified conditions: $U_a = 195$ V; $n = 1400$ rpm, the motor operating current, I, is:

$$n = \frac{U_a - R_a\, I}{K\,\phi} \;\Leftrightarrow\; 1400 = \frac{195 - 0.8\, I}{250/2000} \;\Rightarrow I = 25 \text{ A}$$

When the supply voltage changes suddenly, $(U_a = 195 \text{ V} \rightarrow 155 \text{ V})$, the speed still remains the same (slow dynamics of a mechanical system *vs.* an electrical system), and the current changes to a new value, given by I':

$$n = \frac{U_a - R_a\, I'}{K\,\phi} \;\Leftrightarrow\; 1400 = \frac{155 - 0.8\, I'}{250/2000} \;\Rightarrow I' = -25 \text{ A}$$

With the sudden change in voltage, U_a, the back electromotive force becomes $E' > U_a$, reversing the direction of the current. In other words, a braking process has started, in which the current, I', will follow the evolution of E', being restored to its previous value, $(I = 25 \text{ A})$, once the speed stabilises.

If the electronic variable-speed drive supports bidirectional current, **regenerative braking** occurs: the current is returned to the electrical supply during the braking as long as $E' > U_a$.

Worked solution to an assessment test

1st Test – 29 October 2026

I – A compound-wound DC generator, 100 kW, 400 V, 1500 rpm, with an efficiency of 80 %, is connected in long-shunt with cumulative flux to supply the DC bus of a ship. The armature, shunt-field and series-field winding resistances of the machine are $0.26\,\Omega$, $404\,\Omega$ and $0.10\,\Omega$, respectively.
At a speed of 1200 rpm, the following magnetisation characteristic was obtained:

i [A]	0.0	0.2	0.4	0.6	0.8	0.9	1.0
E_0 [V]	25	200	320	370	392	400	404

a) Draw the electrical circuit diagram of this generator, identifying the machine terminals and the polarities assumed;

b) Determine the value of the shunt-field rheostat required to obtain the rated no-load voltage, assuming that the generator is driven at its rated speed;

c) For the voltage drop due to armature reaction to be zero, $\Delta E = 0$ V, what constructional elements must this machine have? Briefly explain the function of each and indicate where these elements are located inside the machine;

d) Determine the turns ratio $\dfrac{N_d}{N_s}$ of this generator so that its external characteristic has a flat (level) regulation, $\varepsilon = 0\%$, under rated conditions.

II – A shunt-wound DC motor, 40 kW, 300 V, $\eta = 82\,\%$, is used to drive a conveyor belt.

The following additional data are known: armature resistance, $0.23\,\Omega$; voltage drop due to armature reaction at rated current, $\Delta E(I_n) = 49.25$ V; back electromotive force constant, $k\phi = \frac{189}{1000}\left(\frac{V}{\text{rpm}}\right)$; and mechanical and magnetic losses, $p_{(\text{mec+Fe})} = 1799$ W.

Neglect the shunt-field current, I_d, in the calculations, i.e. assume $I_a \approx I$, and calculate:

a) the no-load motor speed and the corresponding loss torque;

b) the speed and useful (shaft) torque at rated operating conditions;

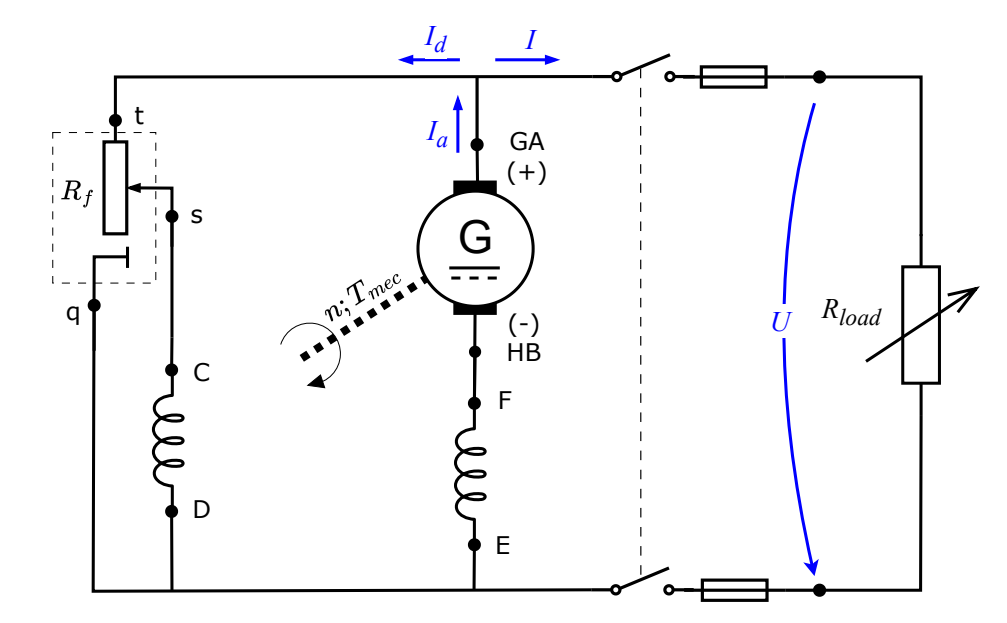
c) Taking all losses into account, represent the power flow diagram of this motor. Include the expressions used to calculate the powers and losses;

d) Sketch, qualitatively and on the same axes, the developed-torque characteristics of this motor without and with armature reaction. Justify.
For the latter characteristic (with ΔE), explain qualitatively the effect of increasing the field rheostat on the characteristic. Justify.

Worked solution:

> *This solution is presented in an explanatory and detailed way, with the aim of helping to understand the reasoning. In a test, the solution may be more concise, provided that the procedure is clear and the results are justified.*

I-a) DC generator with compound excitation, long-shunt connection and cumulative flux:



(Click on the figure to edit or reuse the diagram)

I-b)

$U_0 = U_n = 400\ \text{V}$

$E_0 \approx U_0$, since $\Delta U_r = (R_a + R_s)\, I_d$ is negligible. Thus: $E_0 = 400\ \text{V}$ @ 1500 rpm

Since the magnetic flux depends only on the excitation current, the speeds must be converted in order to consult $E_0(i)$, as follows:

$$E_0 = k\,\phi\,n \implies \begin{cases} 400 = k\,\phi\,1500 \\ E_0^{1200} = k\,\phi\,1200 \end{cases} \implies E_0^{1200} = 400\,\frac{1200}{1500} = 320\ \text{V}$$

At no load, the series flux is negligible, since: $I_d\, N_s \lll I_d\, N_d$

Thus, consulting the magnetisation characteristic, $E_0(i)$, we have:

$$E_0^{1200} = 320\ \text{V} \overset{E_0(i)}{\implies} I_d = 0.4\ \text{A}$$

The field rheostat of the shunt branch, R_f, is obtained as:

$$R_f = \frac{U_0}{I_d} - R_d = \frac{400}{0.4} - 404 = 596\,\Omega$$

I-c)

- **Interpoles** (commutating poles), on the GNA, between the main poles → improve commutation with the brushes fixed on the GNA, preventing sparking.
- **Compensating windings**, in slots in the pole faces of the field poles → cancel the armature MMF under the poles, eliminating the magnetic distortion and the consequent flux reduction.

I-d)

Flat (level) regulation $(\varepsilon = 0\,\%)$ under rated conditions $(U_n, I_n, n_n) \implies U_0 = U(I_n)$ with $U = U_n$, since: $\varepsilon = \dfrac{U_0 - U}{U_n}\,100$

$I_d = 0.4\,\text{A}$, since from part b: $U_0 = U_n$ @ 1500 rpm

Calculation of the rated current from the generator rating plate:

$$I_n = \frac{P_n}{U_n} = \frac{100 \cdot 10^3}{400} = 250\,\text{A}$$

Armature current, I_a: $I_a = I_n + I_d = 250.4\,\text{A}$ or simply $I_a \approx I_n$

Calculation of the EMF on load, considering $\Delta E = 0$ and $\Delta U_{\text{esc}} \approx 0$:

$$E_0 = U + (R_a + R_s)\,I_a = 400 + (0.26 + 0.10)\,250 = 490\,\text{V} \quad @ \quad 1500\,\text{rpm}$$

Determination of the excitation current that represents the total magnetic flux (shunt + series), by consulting the magnetisation characteristic:

$$E_0^{1200} = 490\,\frac{1200}{1500} = 392\,\text{V} \overset{E_0(i)}{\implies} I_{\text{ex}} = 0.8\,\text{A}$$

From the sum of the shunt and series MMFs, the total MMF is given by:

$$N_d\,I_{\text{ex}} = N_d\,I_d + N_s\,I_s, \quad \text{with} \quad I_s = I_a \approx I_n$$

Dividing all terms by N_d and substituting, we have:

$$0.8 = 0.4 + \frac{N_s}{N_d}\,250 \implies \frac{N_d}{N_s} = \frac{250}{0.4} = 625$$

II-a)

At no load, $P_u = 0 \implies P_d = P_{\text{mec+Fe}} = 1799\,\text{W}$

Since: $P_d = E'\,I_{a0}$ and $E' = U - R_a\,I_{a0}$, combining them gives:

$$P_d = U\,I_{a0} - R_a\,I_{a0}^2 \iff R_a\,I_{a0}^2 - U\,I_{a0} + P_d = 0$$

Alternatively, one could also take the approach: $E' = U - R_i\,\dfrac{P_d}{E'}$, and obtain E'.

Substituting the data gives (quadratic formula): $\begin{cases} I_{a0} = 6.0\,\text{A} & \ll I_n \quad \checkmark \\ I_{a0} = 1298.3\,\text{A} & \ggg I_n \quad \times \end{cases}$

Calculation of the no-load speed, n_0, and of the loss torque, T_{loss}:

$$n_0 = \frac{U - R_i\, I_{i0}}{k\,\phi} = \frac{300 - 0.23 \cdot 6.0}{\frac{189}{1000}} = 1580\,\text{rpm}$$

$$T_{\text{loss}} \approx \frac{P_{\text{mec+Fe}}}{\omega_0} = \frac{1799}{\frac{2\,\pi\, n_0}{60}} = \frac{1799}{\frac{2\,\pi\, 1580}{60}} = 10.9\,\text{Nm}$$

II-b)

From the motor rating plate data, the rated current, I_n, is obtained:

$$I_n = \frac{P_n}{\eta\, U_n} = \frac{40 \cdot 10^3}{0.82 \cdot 300} = 162.6\,\text{A}$$

The motor speed under rated conditions is given by:

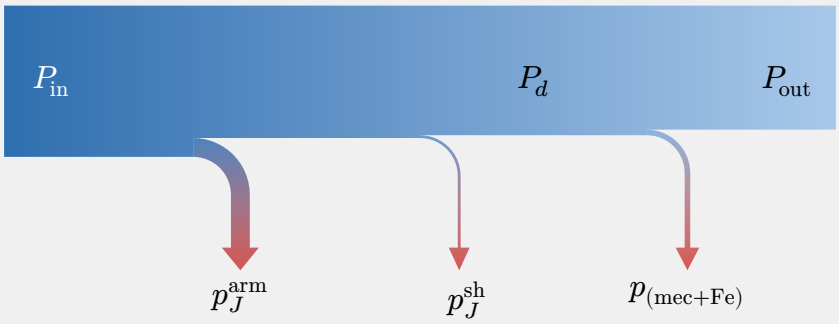
$$n = \frac{U - R_a\, I_n + \Delta E(I_n)}{k\,\phi}, \quad \text{with} \quad \Delta U_{\text{esc}} = 0$$

$$n = \frac{300 - 0.23 \cdot 162.6 + 49.25}{\frac{189}{1000}} = 1650\,\text{rpm}$$

For the calculation of the motor shaft torque under rated conditions, we have:

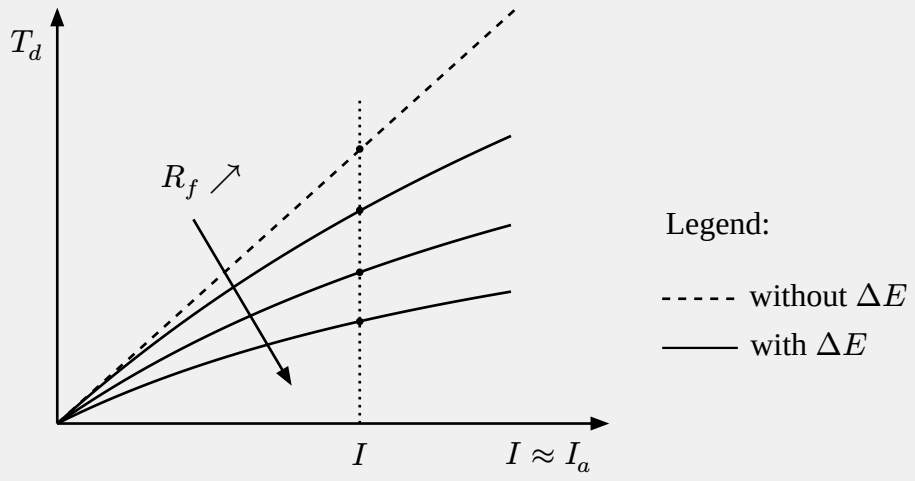
$$T_{\text{out}} = \frac{P_n}{\omega} \quad ; \quad \omega = \frac{2\,\pi\, n}{60} \quad \Longrightarrow \quad T_{\text{out}} = \frac{40000}{\frac{2\,\pi\, 1650}{60}} = 231.5\,\text{Nm}$$

II-c)



Powers	**Losses**
$P_{\text{in}} = U\, I$	$p_J^{\text{arm}} = R_a\, I_a^2$
$P_d = E'\, I_a = T_{\text{mi}}\, \omega$	$p_J^{\text{sh}} = U\, I_d$ or $R_{\text{ex}}\, I_d^2$ or $\frac{U^2}{R_{\text{ex}}}$
$P_{\text{out}} = T_{\text{out}}\, \omega$	$p_{(\text{mec+Fe})} \approx T_{\text{loss}}\, \omega$

II-d) Developed torque characteristic curves, $T_d = f(I)$, with: $\begin{cases} U = \text{constant} \\ R_f = \text{constant} \end{cases}$



Justification of the developed torque characteristics:

$T_d = k\,\phi_0\,I_a$ with $k\,\phi_0 = \text{constant} \implies$ linear characteristic $(\text{without } \Delta E)$

$T_d = k\,\phi\,I_a$ with $k\,\phi < k\,\phi_0$, decreasing as the current I_a increases $\Longrightarrow$ non-linear characteristic $(\text{with } \Delta E)$

or: $T_d = \dfrac{E'_0}{\omega}\,I_a$ $(\text{without } \Delta E)$ *vs.* $T_d = \dfrac{E'}{\omega}\,I_a$ $(\text{with } \Delta E)$, where: $E' < E'_0$

Effect of increasing R_f on the developed torque characteristics:

$R_f \nearrow \implies I_d \searrow \implies \phi \searrow \implies T_d \searrow$

Bibliography

– [Guru & Hiziroğlu, 2003]: Bhag S. Guru, Hüseyin R. Hiziroğlu, Electric Machinery and Transformers, 3rd Ed., Oxford University Press, 2003.

THREE-PHASE SYNCHRONOUS MACHINES

Exercise 1

A three-phase synchronous alternator, 390 kVA, 1250 V, 50 Hz, 750 rpm, delta-connected, gives the following results in the open-circuit (no-load) and short-circuit tests:

I_f (A):	0.0	11.5	15.0	20.0	23.5	29.0	33.5
E_0 (V):	91	1010	1235	1460	1560	1640	1660
I_{sc} (A):	12	144	185	242	284	347	400

The resistance measured at the terminals of the armature winding is $0.144\ \Omega$. Determine:

a) The per-phase resistance of the alternator armature winding, considering a skin-effect correction factor of 1.2;

b) The line voltage, for rated current and an field current of 33.5 A, considering a load power factor of 0.9 lagging;

c) The external characteristic of the three-phase synchronous alternator, with a field current of 33.5 A, for power factors of 0.9 lagging, unity and 0.9 leading;

d) The alternator field current required to supply a three-phase induction motor at a voltage of 1 kV, knowing that the motor develops a power of 150 kW with a power factor of 0.832 and an efficiency of 90 %.

(**Note:** Assume that the synchronous impedance, $\overline{Z}_s$), is the same as that obtained in the previous part)

(Source: Adapted from problem 9 in [Malea & Balaguer, 2004])

Worked solution to Exercise 1: 🔗 interactive *notebook* developed with **Pluto**.jl – julia.

Exercise 2

A four-pole, cylindrical-rotor, three-phase synchronous alternator, with its stator windings connected in delta, is used to supply an isolated electrical network with balanced loads. The alternator is equipped with operational speed and field-current regulators which maintain the frequency and voltage constant, respectively. Nameplate ratings: 72 kVA, 400 V, 50 Hz, $\cos\varphi = 0.83\,(\text{lead})$. The data from the no-load test, short-circuit test and resistance measurements are known:

I_f (A):	0.0	1.0	2.0	3.0	4.0	5.0	6.0	$R_{\text{J-K}} = 100\,\Omega$
E_0 (V):	15	202	370	460	480	491	500	
I_{sc} (A):	2	27	51	76	101	125	150	$R_{\text{U-V}} = 0.6\,\Omega$

a) Draw and determine the parameters of the per-phase equivalent circuit of this machine for a field current of 5 A;

b) Consider a synchronous reactance, $X_s = 7.9\,\Omega/\text{phase}$. Determine the electromotive force and the load angle, $\overline{E}_0 = E_0 \angle \delta$, for the alternator operating under rated conditions;

c) Determine the efficiency and the mechanical torque under rated conditions, knowing that the rotational losses are 1015 W;

d) Suppose that the field-current regulator has failed. Consequently, the field current remains constant at the value specified in part b). Qualitatively analyse what happens to the network voltage if the power factor, $\cos\varphi$, decreases, assuming that the stator current, I, remains constant. Justify, presenting side-by-side phasor diagrams (for the initial condition and after the decrease in $\cos\varphi$). For simplicity, take $R_s = 0\ \Omega$.

Answers:

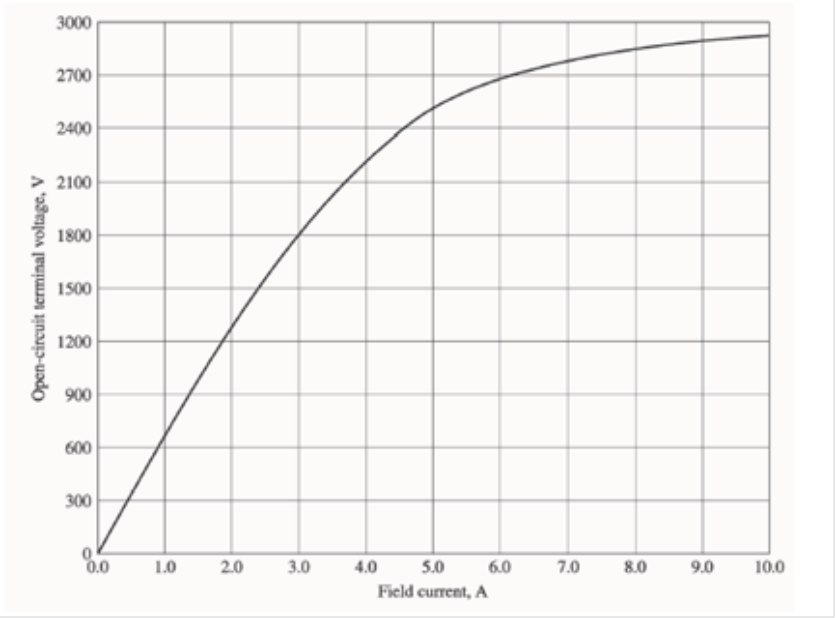
a) $E_0 = 491$ V; $\overline{Z}_s = 0.9 + \text{j}\,6.74\ \Omega$ **b)** $460.4 \angle 66.9°$ V

c) $83.7\,\%$; 448.8 Nm

Exercise 3

A synchronous generator, Y-connected, 2300 V, 1000 kVA, 0.8 lagging power factor, 60 Hz, 2 poles, has a synchronous reactance of 1.1 Ω and an armature resistance of 0.15 Ω. At 60 Hz, the friction and windage losses are 24 kW and the core losses are 18 kW.

The field winding is supplied by a DC voltage of 200 V, the maximum value of I_f being 10 A. The open-circuit test of this alternator is shown in the following figure:



a) What field current is required for the alternator to have a line-to-line voltage to be 2300 V, when the alternator runs at no load?

b) What is the electromotive force (EMF) generated by this machine under rated operating conditions?

c) What field current is required to obtain the rated voltage when the alternator is operating under rated conditions?

d) What power and torque are required to drive this alternator under rated conditions?

e) Obtain the P – Q diagram of this alternator;

f) Under rated conditions, obtain the external characteristic, $U = \mathrm{f}(I)$, for:
$\cos\varphi = 0.8$ lagging; $\cos\varphi = 0.8$ leading; $\cos\varphi = 1$.

g) For an EMF of 2500 V, determine the external characteristic, $U = \mathrm{f}(I)$, for:
$\cos\varphi = 0.8$ lagging; $\cos\varphi = 0.8$ leading; $\cos\varphi = 1$.

(Source: Adapted from problem 5.2 in [Chapman, 2005])

Worked solution to Exercise 3: 🔗 interactive *notebook* developed with **Pluto**.jl – julia.

Exercise 4

A $3\sim$ synchronous machine, 5 kVA, 208 V, 4 poles, 60 Hz, with Y-connected stator, has a negligible stator resistance and a synchronous reactance of 8Ω/phase.
The machine is operated as an alternator connected to a $3\sim$, 208 V – 60 Hz electrical grid.

a) Determine the EMF and the load angle when the machine delivers its rated power (kVA) at a power factor of 0.8 lagging. Draw the voltage phasor diagram for these conditions;

b) Calculate the stator current, power factor and reactive power supplied by the machine if the field current increases by $20\,\%$. Draw the corresponding voltage phasor diagram;

c) With the field current of part a), the mechanical power of the prime mover is gradually increased. What is the steady-state stability limit?
What are the corresponding values of stator current, power factor and reactive power at the condition of maximum power transfer? Draw the phasor diagram.

(Source: Example 6.3 in [Sen, 1989])

Answers:

a) $206.8\angle 25.4°$ V **b)** $17.8\angle -51.6°$ A; $0.62(\text{lag})$; 5.0 kVAr
c) $29.9\angle 30.1°$ A; 0.87 (lead); 5.4 kVAr

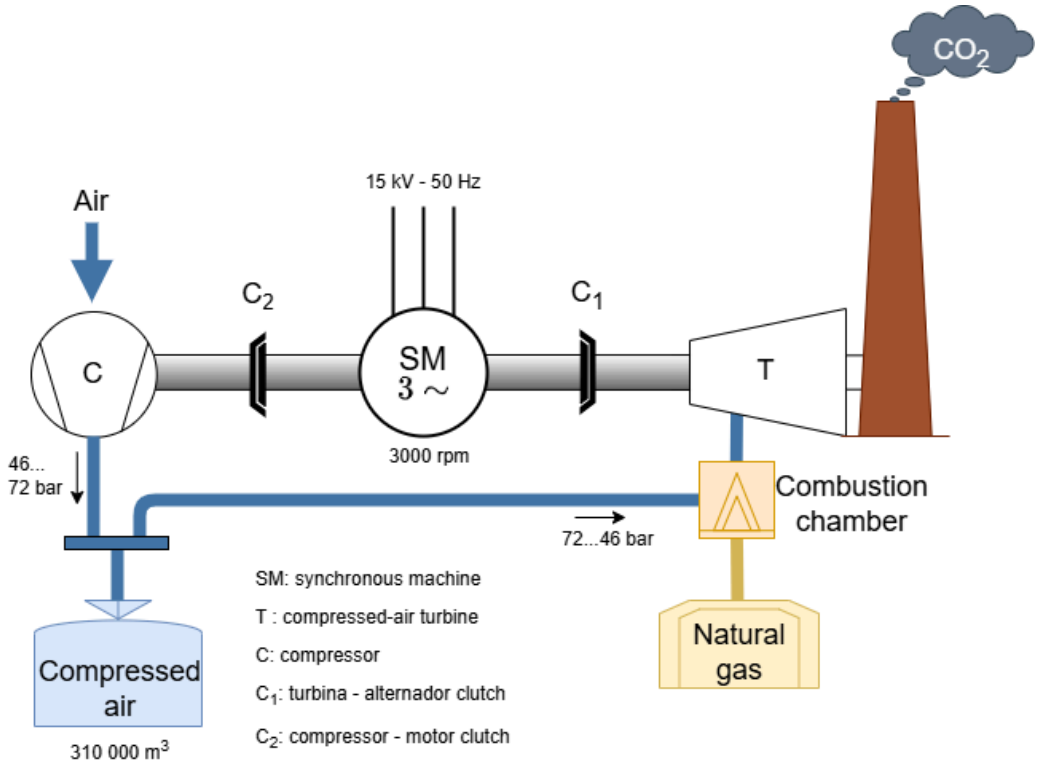
Exercise 5

Consider the Huntorf thermal power plant with compressed-air energy storage (CAES – *Compressed-Air Energy Storage*), which uses a high-speed cylindrical-rotor synchronous machine that can operate as an alternator, a motor or a synchronous compensator.
The plant is connected to the electrical grid at 15 kV – 50 Hz and the synchronous machine has the following parameters: Y-connection, $X_s = 0.34\ \Omega$/phase, $R_a \approx 0\ \Omega$ and rotational losses, $p_{\text{rot}} = 3.2$ MW (assumed constant in all operating modes).

Plant Operating Modes

The plant operates in three modes, enabled by two Synchro-Self-Shifting (SSS) clutches, [SSS gears, 2025], which allow the turbine and the compressor to be coupled to or decoupled from the main shaft:

- **Generation mode (alternator):** Compressed air is heated by burning natural gas and expanded through the turbine, which drives the alternator to generate electricity.
- **Storage mode (motor):** The machine operates as a motor and drives the compressor to store high-pressure air in underground reservoirs (salt caverns).
- **Synchronous compensator mode:** Both clutches are disengaged and the machine runs at no load, focusing on reactive power management to support the grid.



Simplified operating diagram of the Huntorf CAES thermal power plant, Germany. Adapted from [EERA, 2013]

Part 1: Alternator mode (power generation) – Peak period

(In this mode, the turbine clutch is engaged.) The mechanical power supplied by the turbine to the machine shaft is 321 MW. The synchronous machine supplies power to the grid at a lagging power factor of 0.88. Determine:

a) the active power supplied to the electrical grid and the line current;

b) the per phase electromotive force and the load angle;

c) Draw the voltage phasor diagram, indicating the powers on the $Q(\delta), P(\delta)$ and $P(\varphi), Q(\varphi)$ axes.

Part 2: Synchronous motor mode (energy storage) – Off-peak period

(In this mode, the compressor clutch is engaged.) The mechanical power required to drive the compressor is 60 MW. The machine draws power from the grid and is operated over-excited to inject reactive power into the grid, at a power factor of 0.70. Determine:

a) the active power drawn from the electrical grid and the line current;

b) the perphase electromotive force and the load angle;

c) Draw the voltage phasor diagram, indicating the powers on the $Q(\delta), P(\delta)$ and $P(\varphi), Q(\varphi)$ axes.

Part 3: Synchronous compensator mode – Ancillary services

(In this mode, both SSS clutches are disengaged)

a) Starting from the generator mode, as defined in Part 1, describe the procedures required to operate the machine as a **synchronous reactor**;

b) Starting from the motor mode, as defined in Part 2, describe the procedures required to operate the machine as a **synchronous condenser**;

c) If the power station is configured to supply 72 MVAr of inductive reactive power to the electrical grid, determine the required back electromotive force and load angle. Draw the corresponding voltage phasor diagram, indicating the powers on the $Q(\delta), P(\delta)$ and $P(\varphi), Q(\varphi)$ axes.

Worked solution (Part 1):

a)

As an alternator, since the stator Joule losses are neglected $(R \approx 0)$, the active power supplied to the electrical grid, P, is given by:

$$P = P_{\text{turbine}} - p_{\text{rot}} = 321 - 3.2 = 317.8 \text{ MW}$$

In turn, considering the power factor, the line current, I_L, is obtained from:

$$I_L = \frac{P}{\sqrt{3}\, U_L \cos\varphi} = \frac{317.8 \times 10^6}{\sqrt{3} \times 15 \times 10^3 \times 0.88} = 13.90 \text{ kA}$$

b)

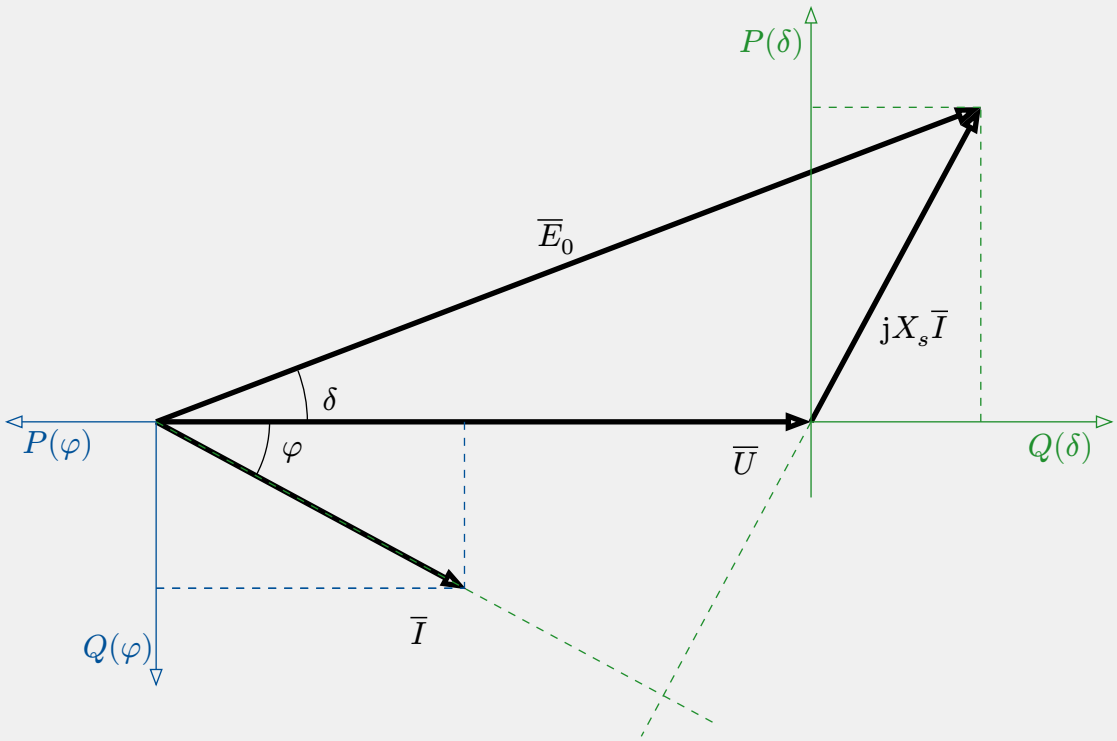
With the stator windings star-connected, the following per-phase quantities apply:

$$\overline{I} = I_L \angle \varphi \quad \text{with:} \quad \varphi = -\arccos(0.88) = -28.4° \quad \text{and} \quad \overline{U} = \frac{U_L}{\sqrt{3}} \angle 0°$$

Thus, $\overline{E}_0$ is given by:

$$\overline{E}_0 = E_0 \angle \delta = \overline{U} + \mathrm{j} X_s \overline{I} = \left(\frac{15 \times 10^3}{\sqrt{3}} \angle 0° \right) + \mathrm{j}\, 0.34 \left(13.90 \times 10^3 \angle -28.4° \right)$$
$$= 11.67 \angle 20.9° \ \text{kV}$$

c)



Worked solution (Part 2):

a)

As a motor, since the stator Joule losses are neglected ($R \approx 0$), the active power drawn from the electrical grid, P, is given by:

$$P = P_{\text{compressor}} + p_{\text{rot}} = 60 + 3.2 = 63.2 \text{ MW}$$

In turn, considering the power factor, the line current, I_L, is obtained from:

$$I_L = \frac{P}{\sqrt{3}\, U_L \cos\varphi} = \frac{63.2 \times 10^6}{\sqrt{3} \times 15 \times 10^3 \times 0.70} = 3475 \text{ A}$$

b)

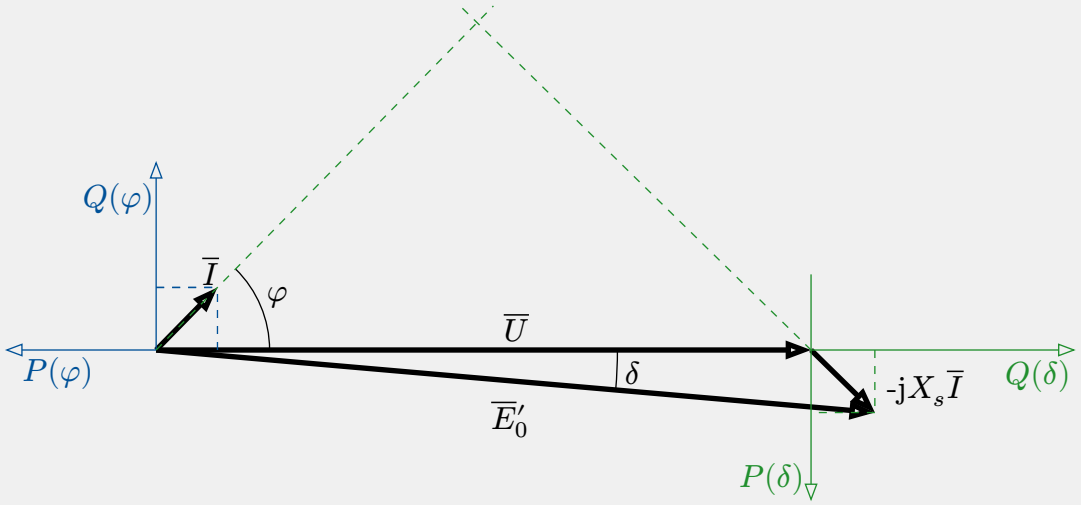
The overexcited motor, $(E'_0 \cos\delta > U)$, requires the current to be capacitive (leading), so:

$$\varphi = \arccos(0.70) = 45.6°$$

Thus, $\overline{E}'_0$ is given by:

$$\overline{E}'_0 = E'_0 \angle\delta = \overline{U} - \mathrm{j}X_s\overline{I} = \left(\frac{15\times10^3}{\sqrt{3}}\angle 0°\right) - \mathrm{j}\,0.34\,(3475\angle 45.6°)$$
$$= 9.54\angle -5.0°\ \text{kV}$$

c)



Worked solution (Part 3):

a)

Starting from operation as an overexcited alternator (Part 1), the machine must be made to "float" on the electric grid, i.e. making: $P \to 0$ and $Q \to 0$.

To do so, the mechanical power of the turbine must be reduced, by reducing the mechanical torque supplied by the turbine to the alternator. As a result:

$$P_{\text{turbine}}\searrow \ \Rightarrow\ \delta\searrow \ \Rightarrow\ P\searrow \text{ and } Q\nearrow \text{(slightly)};\quad \text{until}\quad P=0$$

Simultaneously, the field current of the alternator must be adjusted to reduce the reactive power until it is zero:

$$I_{\text{exc}}\searrow \ \Rightarrow\ Q\searrow \ \text{until}\ \ Q=0$$

As soon as there is no power flow between the synchronous machine and the electrical grid, the turbine-alternator clutch is automatically opened (or decoupled).

Since the synchronous machine remains electrically connected to the grid, it now behaves as a synchronous motor at no load (synchronous compensator). In this mode it draws a low current, essentially to supply its rotational losses, $P = P_{\text{rot}}$.

As a synchronous compensator, to act in synchronous reactor mode, it must absorb an inductive (lagging) current. To do so, the synchronous machine must be under-excited, reducing the field current until the desired reactive power value is obtained.

Thus, the synchronous compensator will draw inductive current, which means that it will be producing capacitive reactive power for the electrical grid.

b)

Starting from operation as an over-excited motor (Part 2), it is enough to reduce the mechanical torque that the compressor exerts on the shaft of the synchronous machine until it is zero, leaving the motor at no load (synchronous compensator). Thus:

$$P_{\text{compressor}} \searrow \;\Rightarrow\; |\delta| \searrow \;\Rightarrow\; |P| \searrow \text{ and } Q \nearrow \text{(slightly)}; \quad \text{until} \quad P = 0$$

As soon as there is no mechanical power transmission (zero mechanical torque) from the synchronous motor to the compressor, the compressor-motor clutch is automatically opened (or decoupled).

For operation as a synchronous condenser, the synchronous compensator must be over-excited (as it already was in Part 2), to draw capacitive current. Thus, it is only necessary to readjust the excitation current to the required reactive power value.

Consequently, and as an example, increasing the excitation current increases the capacitive current drawn by the machine, which allows it to produce more inductive reactive power for the electrical grid:

$$I_{\text{exc}} \nearrow \;\Rightarrow\; E'_0 \nearrow \;\Rightarrow I(\text{cap}) \nearrow \;\Rightarrow Q \nearrow$$

c)

For the synchronous compensator to produce 72 MVAr of inductive reactive power for the grid, it must draw capacitive current, acting as a capacitor bank (synchronous condenser mode). Thus, the apparent power required for the synchronous condenser to operate is given by:

$$\overline{S} = p_{\text{rot}} - \text{j}\, Q = 3.2 - \text{j}\, 72 \;\; \text{MVA}$$

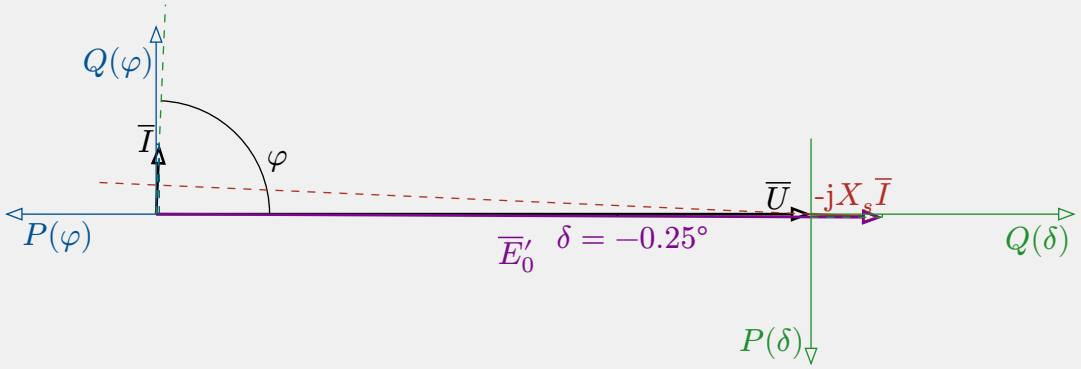
Since: $\overline{S} = 3\,\overline{U}\,\overline{I}^{*}$ with: U, I the voltage and current per phase. Thus:

$$\overline{I} = \left(\frac{\overline{S}}{3\,\overline{U}}\right)^* = \left(\frac{p_{\text{rot}} - \text{j}\,Q}{3\,U\angle 0°}\right)^* = \left(\frac{(3.2 - \text{j}\,72)\times 10^6}{3\,\frac{15\times 10^3}{\sqrt{3}}\angle 0°}\right)^* = 2774\angle 87.5°\ \text{A}$$

Thus, $\overline{E}'_0$ is given by:

$$\overline{E}'_0 = E'_0\angle\delta = \overline{U} - \text{j}X_s\overline{I} = \left(\frac{15\times 10^3}{\sqrt{3}}\angle 0°\right) - \text{j}\,0.34\,(2774\angle 87.5°)$$
$$= 9.6\angle -0.25°\ \text{kV}$$

Phasor diagram:



Exercise 6

The synchronous machine of **Exercise 4** is now used as a synchronous motor supplied from a 3 ∼, 208 V – 60 Hz network. The field current is adjusted to obtain unity power factor when the machine draws 3 kW from the grid.

a) Determine the back electromotive force (back EMF) and the load angle. Draw the voltage phasor diagram under these conditions;

b) Determine the maximum torque the motor can develop if the field current is kept constant and the load applied to the shaft is gradually increased.

(Source: Example 6.4 in [Sen, 1989])

Answers:

a) $137.3\angle -29.0°$ V **b)** 32.8 Nm

Exercise 7

Consider a synchronous alternator with $X_s = 4\ \Omega$ and $R \approx 0\ \Omega$, operating on an infinite-power bus: $U = 200\ \text{V/phase}$ and $f = 50\ \text{Hz}$.

a) Draw the phasor diagrams corresponding to the combinations of the following active and reactive powers:

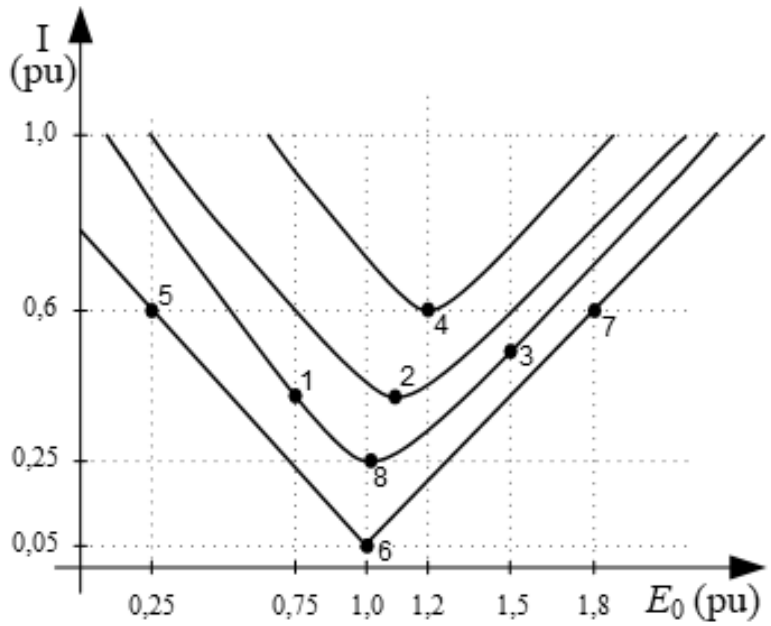
0 5 10 15 20 kW, kVAr

b) Plot the V-curves, also known as Mordey curves, for the required power levels.

Worked solution to Exercise 7: 🔗 interactive *notebook* developed with **Pluto**.jl – julia.

Exercise 8

Consider a cylindrical-rotor synchronous machine, 2 poles, 20 MVA, 6 kV−50 Hz, $\cos\varphi = 0.8$ (lagging), Y-connected stator windings of negligible resistance and $X_s = 1.1$ pu. The synchronous machine is connected to a 6 kV−50 Hz electrical grid and has the operating map represented by the V-curves in the following figure:



Determine in per-unit (pu) values:

a) the active and apparent power at operating point 4 and the corresponding load angle;

b) Which points form an optimum-excitation line? Justify;

c) at points 1 and 3, the active and reactive powers and the load angles, as an alternator. Present side by side the voltage phasor diagrams of points 1 and 3, indicating the respective powers;

d) the rotational losses of the machine. Justify.

e) the mechanical torque corresponding to point 8, operating as a synchronous motor. Present the corresponding voltage phasor diagram;

f) Select, with justification, the point corresponding to synchronous condenser operation and draw its voltage phasor diagram.

Answers:

a) $Q_4 = 0$ pu; $P_4 = \pm 0.6$ pu; $\delta_4 = \pm 33.4°$ **b)** optimum excitation: $6 - 8 - 2 - 4$

c) $P_1 = P_3 = 0.25$ pu; $\delta_1 = 21.5°$; $\delta_3 = 10.6°$; $Q_1 = 0.27$ pu (leading); $Q_3 = 0.43$ pu (lagging)

d) $p_{\text{rot}} = 0.05$ pu **e)** $T_{\text{mec}} = 0.2$ pu **f)** synchronous condenser: 7

Exercise 9

A three-phase synchronous machine, 5 MVA, 11 kV, with Y-connected stator windings, has a synchronous reactance of 10 Ω/phase and a negligible armature resistance. The machine is connected to an 11 kV$-$60 Hz busbar and operates as a synchronous compensator. Neglect the mechanical losses.

a) Determine the stator current at the optimum-excitation point. Draw the phasor diagram;

b) Determine the stator current and the power factor if the field current is increased by 50 %. Draw the phasor diagram;

c) Likewise, for a decrease of 50 %. Draw the phasor diagram.

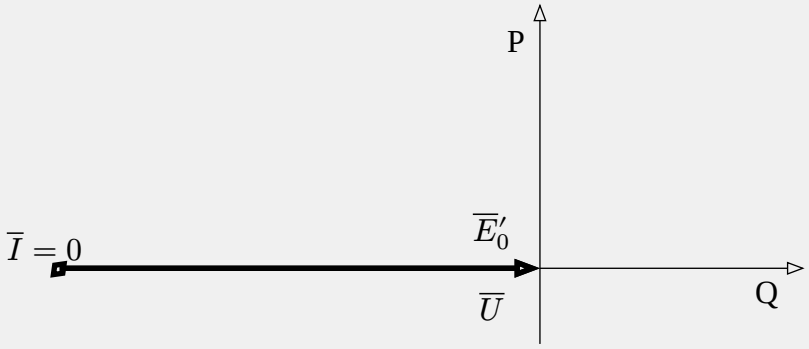
(Source: Example 6.6 in [Sen, 1989])

Worked solution:

a)

With no losses to consider and since this is a synchronous compensator $(P = 0)$, the current will be zero at the optimum excitation point $(Q = 0, \quad \text{or} \quad \cos\varphi = 1)$, hence $I = 0$ A.

Phasor diagram:



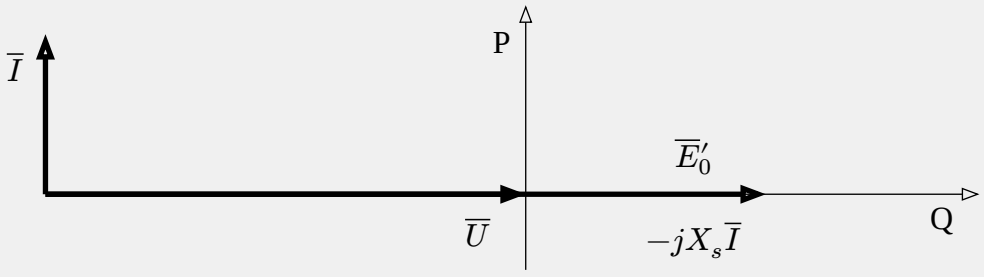
b)

Assuming a linear magnetic circuit (the magnetisation characteristic being unknown), a $50\,\%$ increase in the excitation current produces the same increase in the electromotive force, $\overline{E}_0$. Thus, starting from the previous situation, $\left(\overline{E}'_0 = \overline{U}\right)$, we have:

$E'_0 = 1.5\,U$

As a synchronous compensator, neglecting losses, the load angle is zero, $\delta = 0°$.

Phasor diagram:



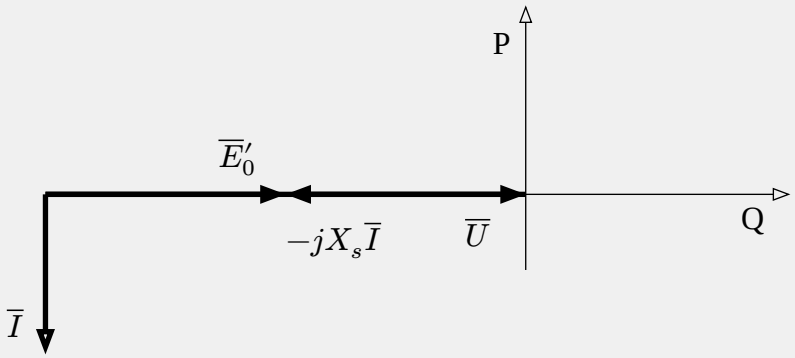
$$\overline{E}'_0 = \overline{U} - \mathrm{j}\,X_s\,\overline{I} \quad \Leftrightarrow \quad \overline{I} = \frac{\overline{E}'_0 - \overline{U}}{-\mathrm{j}\,X_s} \quad \Leftrightarrow$$

$$\Leftrightarrow \quad \overline{I} = \frac{\left(\dfrac{1.5 \times 11 \times 10^3}{\sqrt{3}}\angle 0°\right) - \left(\dfrac{11 \times 10^3}{\sqrt{3}}\angle 0°\right)}{-\mathrm{j}\,10} = 317.5\angle 90° \ \text{A}$$

c)

Using similar reasoning to the previous part, we have: $E'_0 = 0.5\,U$ and $\delta = 0°$.

Phasor diagram:



$$\overline{E}'_0 = \overline{U} - \mathrm{j}\,X_s\,\overline{I} \quad \Leftrightarrow \quad \overline{I} = \frac{\overline{E}'_0 - \overline{U}}{-\mathrm{j}\,X_s} \quad \Leftrightarrow \quad \overline{I} = 317.5 \angle -90° \text{ A}$$

Exercise 10

Consider a three-phase synchronous alternator, 20 MVA, 11 kV, with star-connected stator windings of negligible resistance. The synchronous alternator has the following magnetisation characteristic at rated speed:

I_f (A)	60	80	100	120	140	160	180	200	220	240
E_0 (V)	7772	8827	9611	10214	10694	11085	11409	11683	11917	12119

The alternator is connected to an 11 kV infinite-power busbar, supplying a power of 20 MW at unity power factor. For an field current of 200 A, determine:

a) The synchronous reactance of the alternator;

Consider a second three-phase synchronous alternator, with identical rated characteristics, connected in parallel with the first alternator to the 11 kV infinite-power busbar and carrying the same field current. If the two alternators together supply a power of 36 MW, shared equally between them, at a power factor of 0.9 lagging, calculate:

b) The induced EMF in each three-phase synchronous alternator and the total current supplied to the electrical grid;

If the per phase EMF of one of the alternators operating in parallel is reduced by $10\,\%$, while the mechanical torque applied by the turbine to each alternator and the overall power factor are kept constant, determine:

c) The induced EMF in the other three-phase synchronous alternator required to meet the stated operating conditions;

d) The currents supplied to the grid and the power factor of each alternator.

(Source: Adapted from problem 10 in [Malea & Balaguer, 2004])

Answers:

a) $2.16\ \Omega$; **b)** $7618\angle 15.5°\ \text{V};\ 2099\angle -25.8°\ \text{A}$; **c)** $8385\angle 14.1°\ \text{V}$

d) $949\ \text{A};\ 0.996\ (\text{lag});\quad 1254\ \text{A};\ 0.753\ (\text{lag})$

Exercise 11

A salient-pole synchronous alternator, 12 MVA, delta-connected, $\eta = 91\%$, is part of a cogeneration process at a pulp and paper mill and is connected to an $13.8\ \text{kV}-50\ \text{Hz}$ infinite-power busbar. The following parameters are known: $X_d = 34\Omega/\text{phase}$; $X_q = 16\ \Omega/\text{phase}$ and $R \approx 0\ \Omega$.

a) Qualitatively draw the voltage phasor diagram at full load and $\cos\varphi = 0.81$ (leading), showing how the direct and quadrature axes are determined and the corresponding current components along these axes;

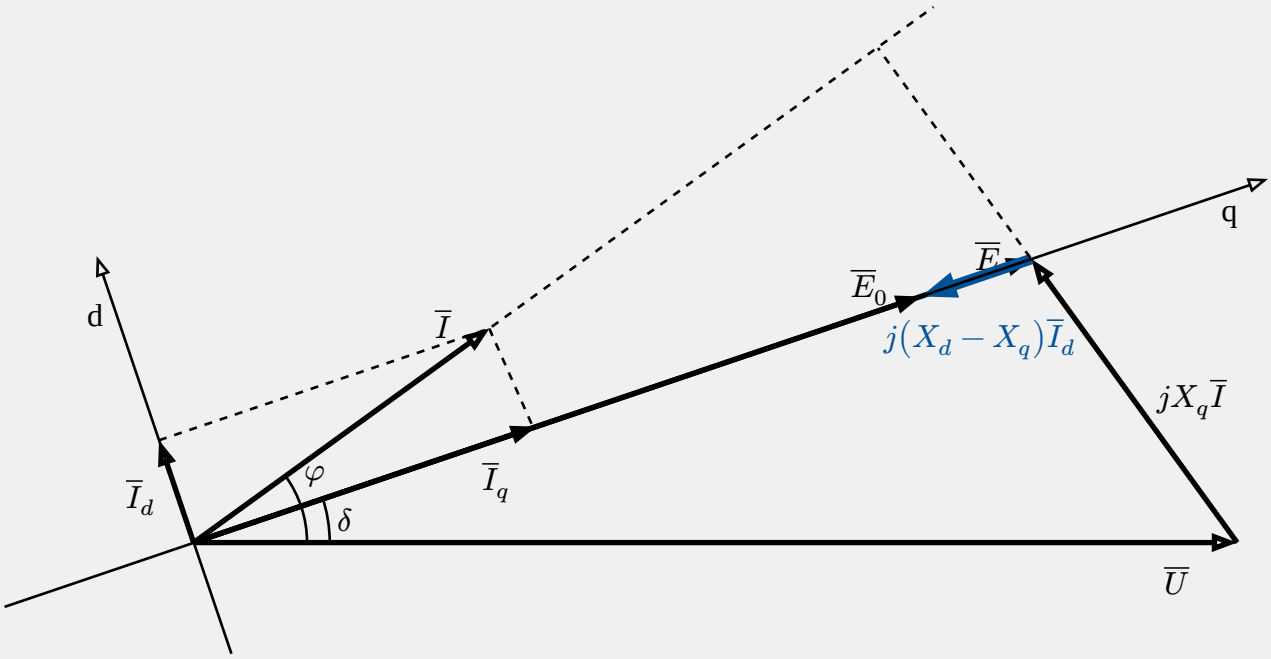
b) Determine the electromotive force phasor under these operating conditions;

c) Keeping the excitation current constant, determine the static stability limit of this alternator and the corresponding developed power;

d) Qualitatively:
   - How to proceed to set this synchronous machine to operate as a synchronous condenser in this installation, starting from the operating point described;
   - Present the phasor diagram of the new operating point of the synchronous machine.

Worked solution:

a)



b)

Voltage and current phasors per phase (equivalent circuit), $\overline{U}_f$ and $\overline{I}_f$:

$$I_n = \frac{S_n}{\sqrt{3}U_n} = \frac{12 \times 10^6}{\sqrt{3} \cdot 13.8 \times 10^3} = 502 \text{ A} \qquad \text{phase angle: } \varphi = \arccos(0.81) = 35.9°$$

Stator in Δ: $\quad U_f = U_n \; ; \quad I_f = \dfrac{I_n}{\sqrt{3}}$

Phasors: $\quad \overline{U}_\text{f} = 13.8 \angle 0° \text{ kV} \qquad \overline{I}_\text{f} = \dfrac{502}{\sqrt{3}} \angle 35.9° \text{ A}$

Effective EMF phasor, $\overline{E}$:

$$\overline{E} = \overline{U}_f + jX_q\overline{I}_f = (13.8 \times 10^3 \angle 0°) + j16\left(\frac{502}{\sqrt{3}} \angle 35.9°\right) = 11.7 \angle 18.7° \text{ kV}$$

Current phasor along the direct axis, $\overline{I}_d$:

$$\overline{I}_d = I \sin(\varphi - \delta) \angle(\delta + 90°) = \frac{502}{\sqrt{3}} \sin(35.9° - 18.7°) \angle(18.7° + 90°)$$

$$\overline{I}_d = 85.7 \angle 108.7° \text{ A}$$

EMF phasor, $\overline{E}_0$:

$$\overline{E}_0 = \overline{E} + j(X_d - X_q)\overline{I}_d = (11.7 \times 10^3 \angle 18.7°) + j(34 - 16)(85.7 \angle 108.7°)$$

$$\overline{E}_0 = 10.16 \angle 18.7° \text{ kV}$$

c)

$$\text{Static stability limit: } \Rightarrow \frac{\mathrm{d}P_d}{\mathrm{d}\delta} = 0 \Leftrightarrow$$

$$\frac{3UE_0}{X_d}\cos\delta_{\text{lim}} + \frac{3U^2\left(X_d - X_q\right)}{X_d X_q}\cos(2\delta_{\text{lim}}) = 0 \Leftrightarrow$$

$$\frac{3 \cdot 13.8 \times 10^3 \cdot 10.16 \times 10^3}{34}\cos\delta_{\text{lim}} + \frac{3\left(13.8 \times 10^3\right)^2(34-16)}{34 \cdot 16}\cos(2\delta_{\text{lim}}) = 0 \Rightarrow$$

$$\delta_{\text{lim}} = 55.8°$$

With: $I_{\text{exc}} = \text{constant} \Rightarrow E_0 = \text{constant}$

Substituting into $P_d(\delta)$ with $\delta = \delta_{\text{lim}}$:

$$P_d^{\max} = \frac{3UE_0}{X_d}\sin\delta_{\text{lim}} + \frac{3U^2\left(X_d - X_q\right)}{2X_d X_q}\sin(2\delta_{\text{lim}})$$

$$= \frac{3 \cdot 13.8 \times 10^3 \cdot 10.16 \times 10^3}{34}\sin(55.8°) + \frac{3\left(13.8 \times 10^3\right)^2(34-16)}{2 \cdot 34 \cdot 16}\sin(2 \cdot 55.8°)$$

$$\approx 19 \text{ MW}$$

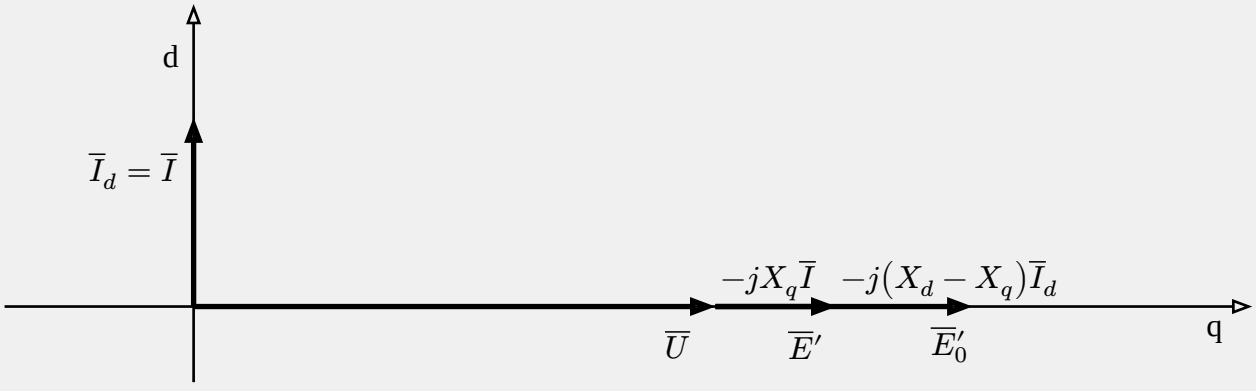
d) Procedure for operating as a synchronous condenser:

1st : (T_{mec} ↘ until $\delta = 0$) and (I_{exc} ↗ until $\overline{E}_0 = \overline{U}$) ⇒ machine “floating” on the network, i.e. $(Q = 0, P = 0)$

2nd : Decouple the turbine from the alternator ⇒ synchronous motor at no load

3rd : (I_{exc} ↗ until Q desired), overexcited mode ⇒ synchronous condenser

Voltage phasor diagram of the salient-pole synchronous condenser:



Exercise 12

A three-phase synchronous motor with four salient poles and a star-connected stator is operating on a 208 V infinite bus, at $\cos\varphi = 0.8$ (lagging), drawing 40 A.

It is known that: $f = 50$ Hz; $X_d = 2.7\ \Omega$/phase; $X_q = 1.7\ \Omega$/phase; $R = 0\ \Omega$.

a) Qualitatively draw the corresponding voltage phasor diagram, showing how the direct and quadrature axes are determined and the current components along these axes;

b) Determine the back electromotive force phasor, $\overline{E}'_0$, under the stated operating conditions;

c) Calculate the maximum torque developed by this machine;

d) Qualitatively draw the developed-torque characteristic of this motor.

Answers:

b) $94.5 \angle -34.5°$ V; **c)** 95.8 Nm

Exercise 13

A $3\sim$ salient-pole synchronous machine, 50 MVA, 11 kV−60 Hz, with Y-connected stator windings, has reactances: $X_d = 0.8$ pu and $X_q = 0.4$ pu. The machine operates as a synchronous motor at full load with a power factor of 0.8 lagging. The mechanical losses are 0.15 pu.
Neglect the armature Joule losses.

a) Determine X_d and X_q in Ω;

b) Determine the EMF in pu;

c) Determine the powers developed (in pu) due to the excitation EMF and due to the reluctance effect of the rotor;

d) If the field current is reduced to zero, does the machine remain in synchronism? Justify;

e) If the shaft load is removed and the field current is reduced to zero, determine the value of the stator current (in pu) and the power factor. Draw the machine voltage phasor diagram for this operating condition.

Worked solution to Exercise 13: 🔗 interactive *notebook* developed with **Pluto**.jl – julia.

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Electrical Machine Transients

Exercise 1

Analyse separately the following two three-phase synchronous alternators, each star-connected to an infinite bus of 6 kV – 50 Hz. Each alternator initially supplies 1 MW at $\cos\varphi = 1$.

	Alternator 1	**Alternator 2**
Synchronous machine:	cylindrical rotor	salient poles
Rated power:	5 MVA	5 MVA
Reactances/phase:	$X_s = 12.75\,\Omega$	$X_d = 12.75\,\Omega$; $X_q = \dfrac{X_d}{2}$
Losses:	negligible	negligible
EMF/phase:	$E_0 = 3.675$ kV	$E_0 = 3.675$ kV

Suppose that a fault occurs in the turbine governor, causing the mechanical power, P_{mec}, to increase suddenly from 1 MW to a new constant value, $P_{\text{mec}} > 1$ MW. The infinite bus keeps the frequency and the voltage constant, absorbing the excess active power and any variation in reactive power.

Using the equal-area criterion, determine the maximum permissible disturbance for each alternator while keeping the synchronous machine stable. Discuss the differences between the alternators.

Worked solution to Exercise 1: 🔗 interactive *notebook* developed with **Pluto**.jl – julia.

Exercise 2

A three-phase cylindrical-rotor synchronous motor, 800 kW, $X_s = 1.2$ pu, connected to an infinite-power electrical grid at 6.6 kV – 50 Hz, drives the main line of a polypropylene film extruder.

In steady state, the motor draws a power of 560 kW at unity power factor ($Q \approx 0$). Neglect Joule and rotational losses.

Suddenly the film tears in the stretching oven, causing the mechanical load on the motor shaft to drop instantaneously from 560 kW to 0 kW (the motor is left at no load).

Assuming that the field current remains constant, determine:

a) the initial load angle, δ_0;

b) the maximum angle reached by the rotor during the transient, $\delta_{\text{máx}}$;

c) Justify whether the motor remains in synchronism or loses it, using the equal-area criterion.

Note: It is not necessary to calculate the areas numerically. Obtain the answer by simple inspection of the accelerating and decelerating areas, without any integration (use only the physical interpretation).

Answers:

a) $-40.0°$; **b)** $40.0°$

Exercise 3

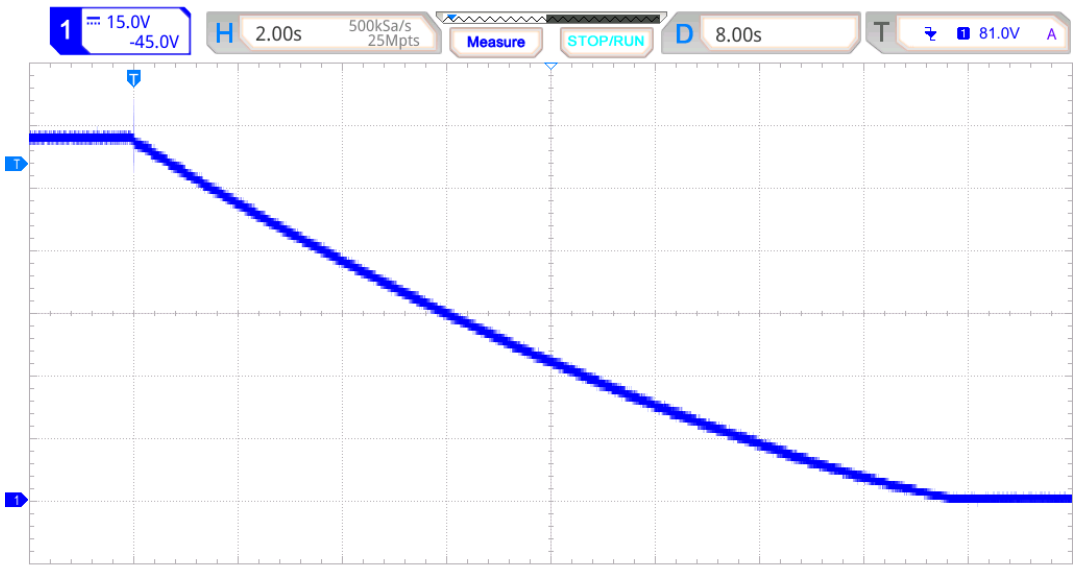
Consider a separately excited DC motor on which two tests were carried out:

- No-load test to separate the mechanical and magnetic losses;
- Deceleration (run-down) test to determine the mechanical parameters.

No-load test (armature voltage and current at constant speed = 1500 rpm)

(V)	220	210	200	190	180	170	160	151	140	130	121	110	100	91
(mA)	2720	2730	2740	2780	2816	2870	3036	3080	3180	3353	3492	3730	4010	4335

Deceleration test (vertical scale: 1500 rpm <> 5.8 div):



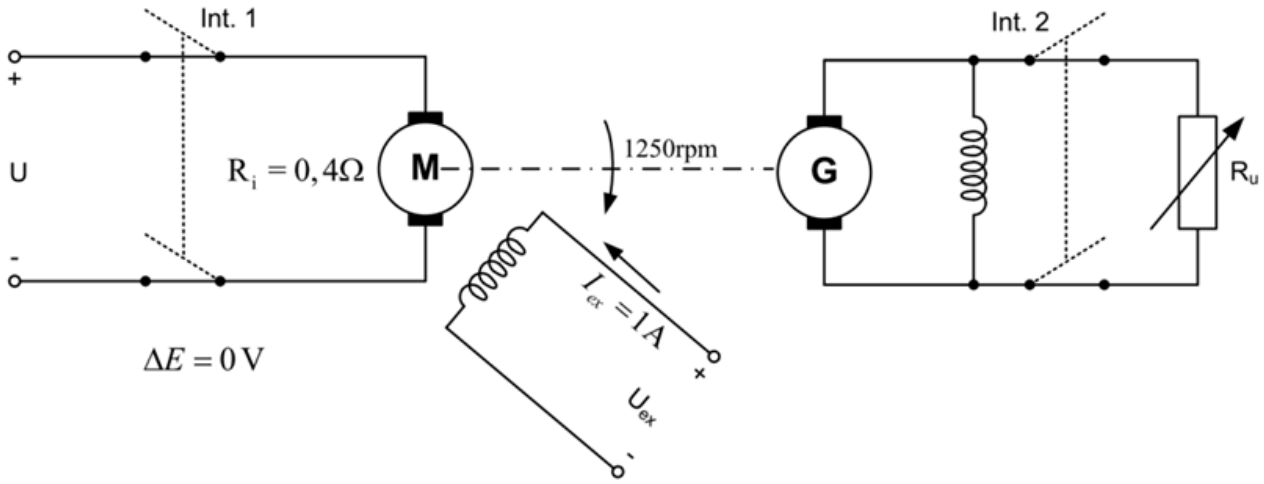
a) Determine the mechanical losses of this motor, given that the armature resistance is $R_a = 0.834\,\Omega$;

b) Determine the mechanical parameters of this motor (moment of inertia, viscous and static friction coefficients).

Worked solution to Exercise 3: 🔗 interactive *notebook* developed with **Pluto**.jl – julia.

Exercise 4

Consider two DC machines connected according to the electrical diagram in the following figure:



The shunt-excited generator, **G**, supplies the load resistance, **Ru**, and is driven at 1250 rpm by a separately excited motor, **M**, which has the following magnetisation characteristic:

i (A)	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	$n_{\text{mag}} = 1000$ rpm
E (V)	15	75	130	170	200	220	235	250	

a) Write down the equations modelling the dynamic operation of the DC motor;

b) Consider the motor supplied at a voltage of 295 V and a current of 50 A. Assuming that the flux is constant, calculate the load torque applied to the motor under these conditions, given that the system (motor + load) has the following mechanical parameters:

$J = 0.4$ kgm^2 $K_d = 0.07$ Nm/rad.s^{-1} $K_e = 3.5$ Nm

c) Suppose that the motor is operating under the following conditions:

$T_d = 100$ Nm $T_l = 80$ Nm $n = 2250$ rpm

Determine the value of $\frac{\mathrm{d}\,\omega}{\mathrm{d}\,t}$ when: 1) → switch 1 is opened;
2) → switch 2 is opened.

Answers:

b) 92.3 Nm; **c-1)** $-250\ \text{rads}^{-2}$; **c-2)** $200\ \text{rads}^{-2}$



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